



CLOUD TECHNOLOGIES

Shivaleela B H
Computer Applications



CLOUD TECHNOLOGIES

Introduction

Shivaleela B H
Computer Applications

At the end of this session, students will be able to:

- Define Cloud Computing.
- Explain why cloud computing has become popular.
- Describe the evolution from traditional computing to cloud computing.
- Identify the basic characteristics of cloud computing.
- Recognize common cloud applications used in daily life.
- Understand how cloud computing benefits individuals and businesses.

CLOUD TECHNOLOGIES

Introduction To Cloud Technologies

- Cloud computing has transformed the way organizations store data, develop applications, and deliver services. Instead of relying on local computers or physical servers, users can access computing resources over the Internet. This technology has become the foundation of modern digital services such as online learning, video streaming, e-commerce, and banking.
- Cloud computing is one of the most demanded technologies in the IT industry.
- We are using cloud services daily, often without realizing it.

CLOUD TECHNOLOGIES

Introduction To Cloud Technologies

- **Notebook:** Allows users to access cloud applications and files from a laptop.
- **Mobile:** Enables users to use cloud services anytime and anywhere through smartphones.
- **PC:** Connects to the cloud for data storage, software, and computing resources.
- **Server:** Hosts applications and processes user requests in the cloud.

CLOUD TECHNOLOGIES

Introduction To Cloud Technologies

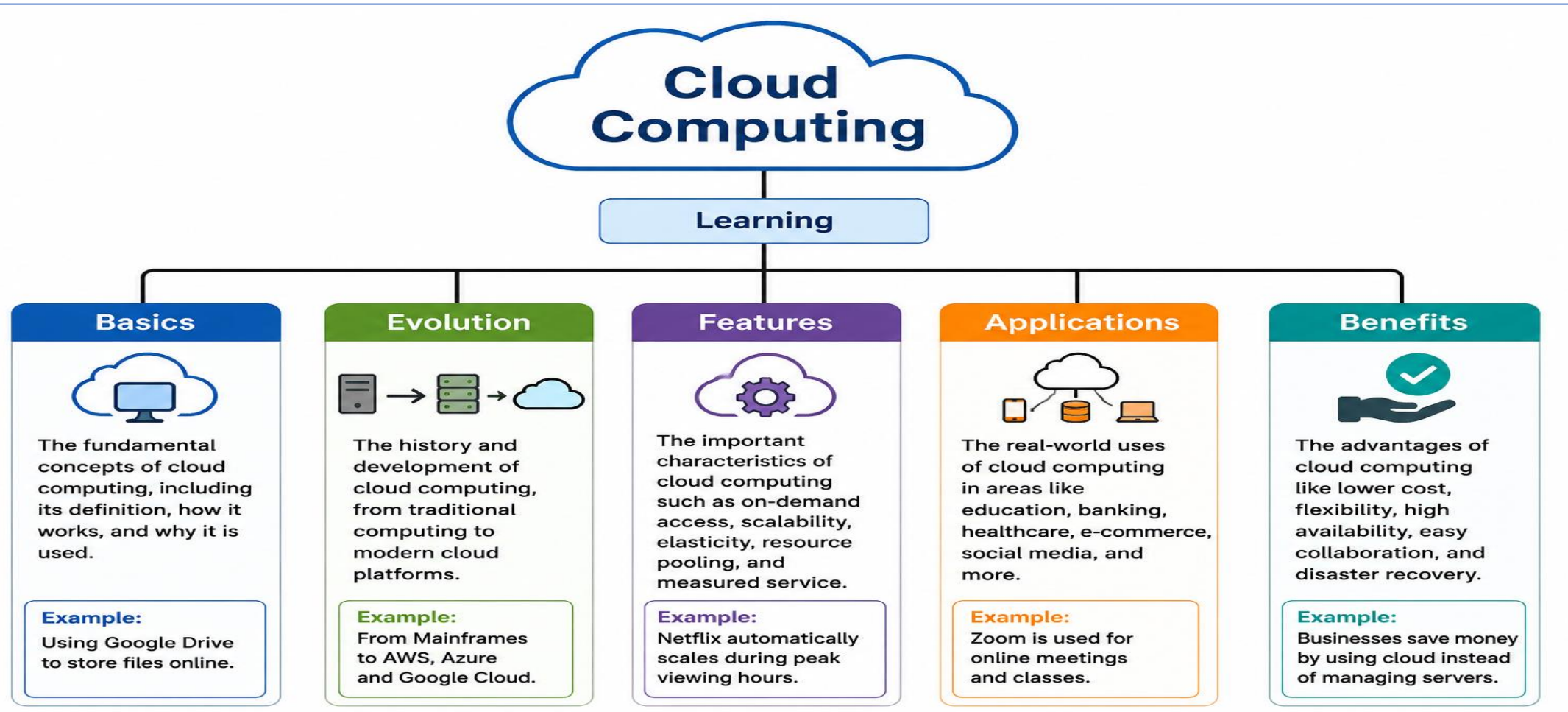
- **Database:** Stores and manages data securely for cloud applications.
- **Network:** Connects users and cloud resources through the Internet.



CLOUD TECHNOLOGIES

Introduction To Cloud Technologies

Overview



CLOUD TECHNOLOGIES

Cloud Computing

Cloud Computing is the delivery of computing services such as servers, storage, databases, software, and networking **over the Internet**("the cloud").

Instead of buying and maintaining expensive hardware, users access these resources from cloud providers on a **pay-as-you-use** basis.

It allows users to access applications and data **anytime, anywhere** through an Internet connection.

Cloud computing offers **flexibility, scalability, and cost savings** for businesses and individuals.

CLOUD TECHNOLOGIES

Cloud Computing

- According to **NIST**(National Institute of Standards and Technology), it **provides on-demand** access to a shared pool of configurable computing resources with minimal effort.
- **Ex:** Cloud computing is similar to using electricity. We do not own the power plant—we simply use electricity when needed and pay based on consumption. Cloud computing works similarly by providing computing resources on demand.
- Google Drive stores documents online instead of on PC.

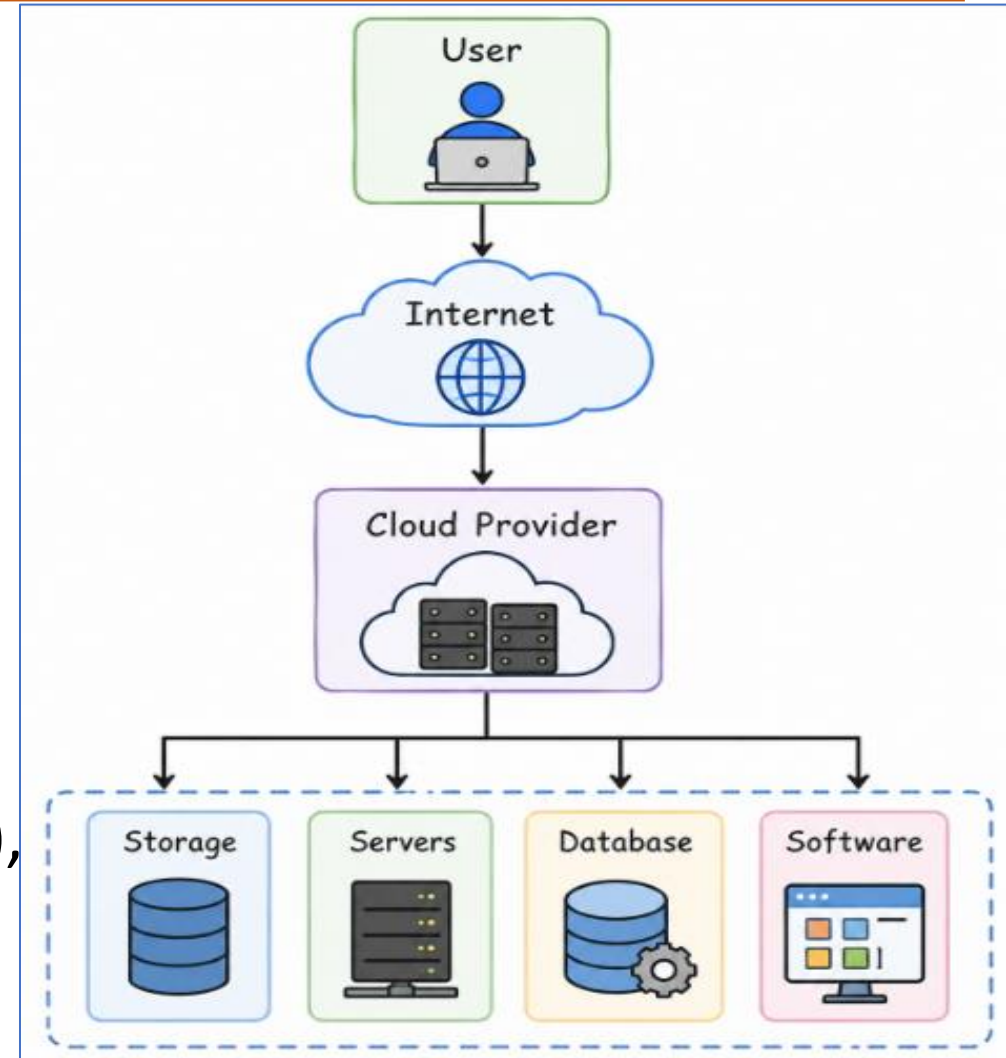
CLOUD TECHNOLOGIES

Cloud Computing

Cloud Computing Architecture Diagram:

User accesses cloud services through the internet. The cloud provider delivers resources such as storage, servers, databases, and software on demand.

Cloud Providers : Amazon Web Services (AWS),
Microsoft Azure, Google Cloud (GCP)



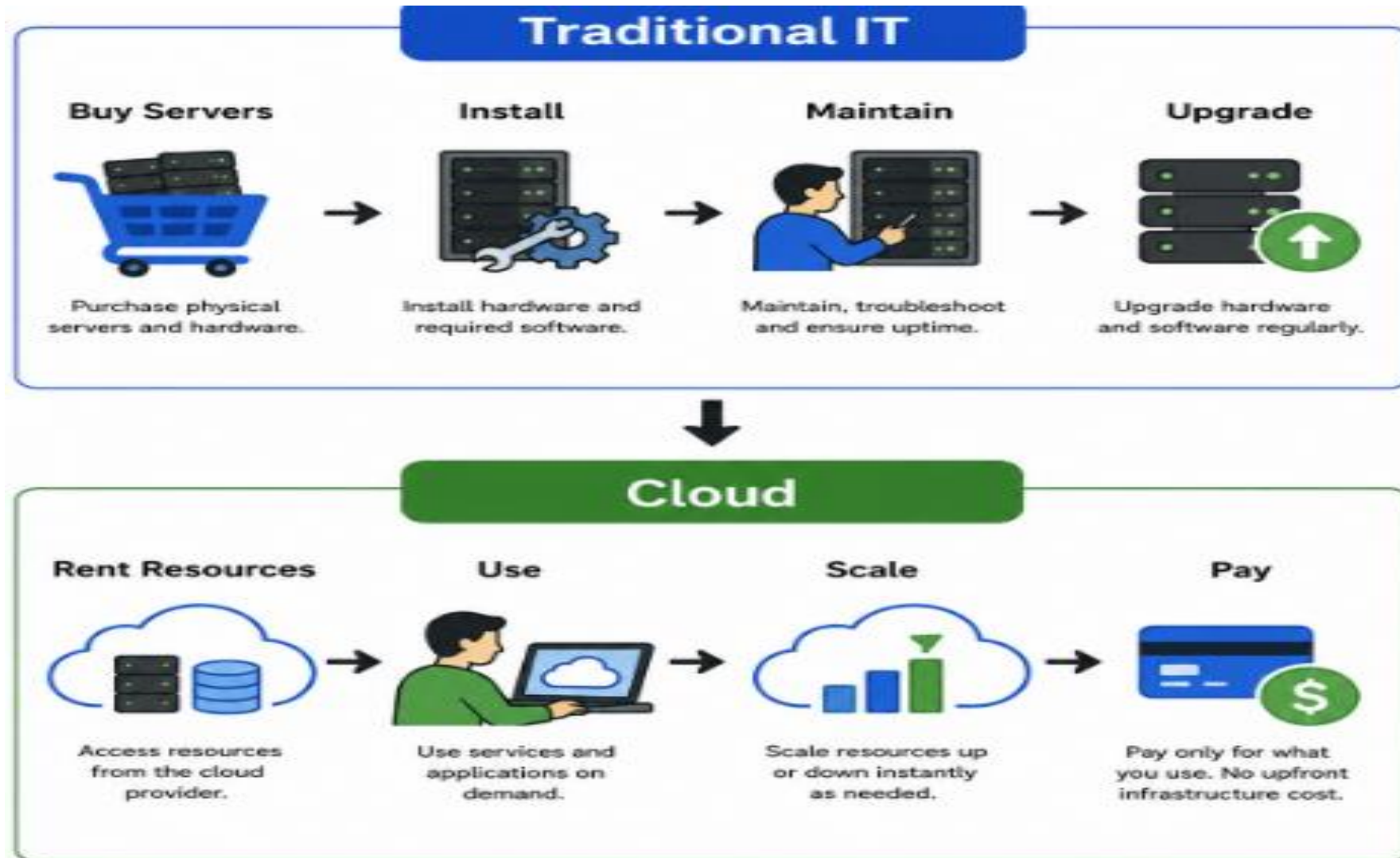
CLOUD TECHNOLOGIES

Need Of Cloud Computing

- **Traditional computing** systems require organizations to purchase servers, software licenses, networking equipment, and storage devices. These systems involve significant installation, maintenance, upgrades, and operational costs.
- **Cloud computing** addresses these challenges by offering scalable and cost-effective computing resources over the Internet. Organizations can quickly increase or decrease resources depending on demand, ensuring efficient utilization and reduced expenses.

CLOUD TECHNOLOGIES

Need Of Cloud Computing



CLOUD TECHNOLOGIES

Need Of Cloud Computing

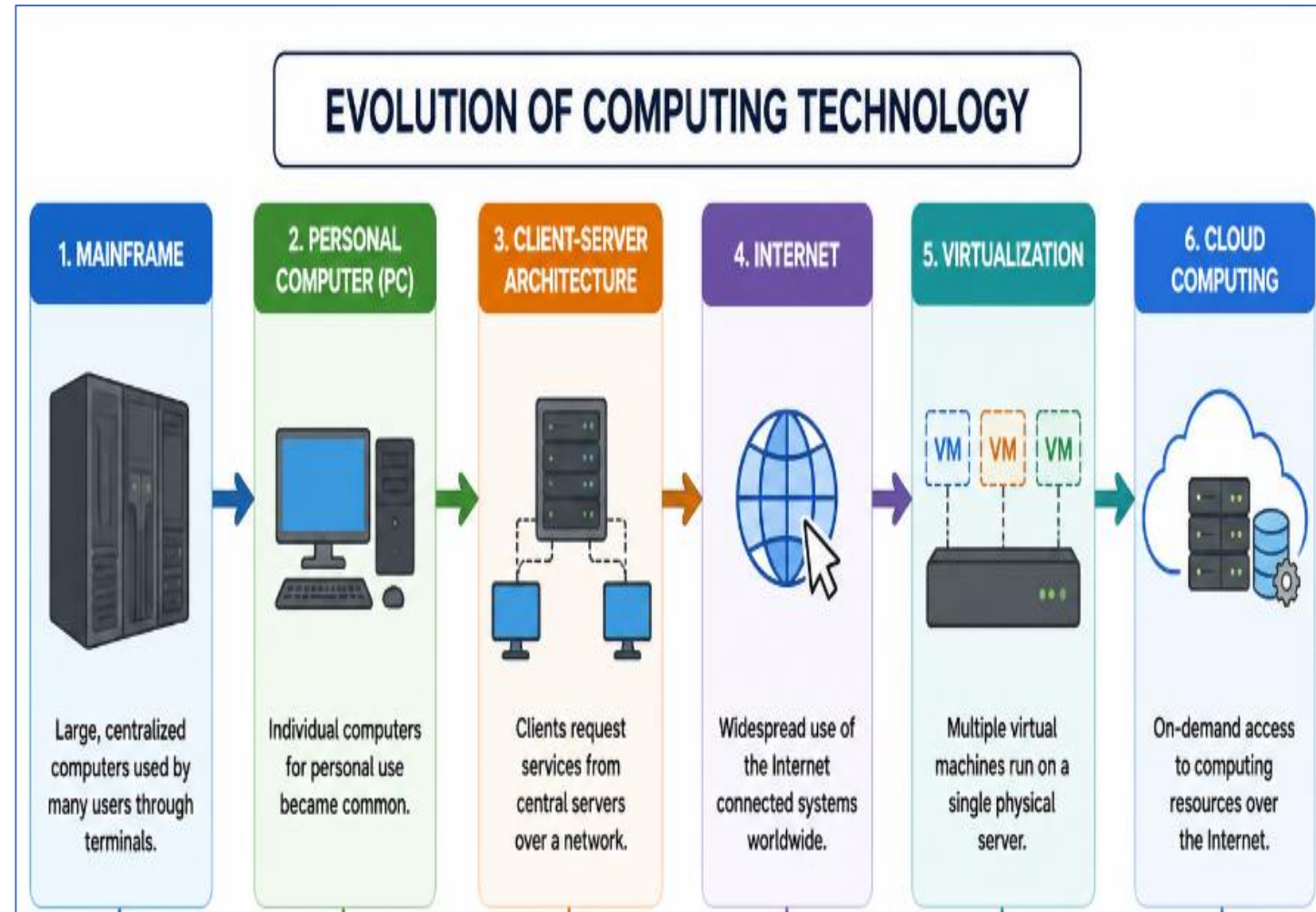
- The increasing use of mobile devices, remote work, digital transformation, and global collaboration has further accelerated the adoption of cloud computing across industries.
- **Cloud Computing Reduce infrastructure cost , Faster deployment , Remote accessibility Scalability , Business continuity , Disaster recovery**
- **Ex:** A startup hosting its website on AWS/Google Cloud instead of purchasing physical servers.

CLOUD TECHNOLOGIES

Evolution of Computing

Cloud Computing Evolution

- Mainframe Computing
- Personal Computers
- Client-Server Architecture
- Internet Computing
- Virtualization
- Cloud Computing



CLOUD TECHNOLOGIES

Characteristics of Cloud Computing

- Cloud computing offers several unique characteristics that distinguish it from traditional computing systems.

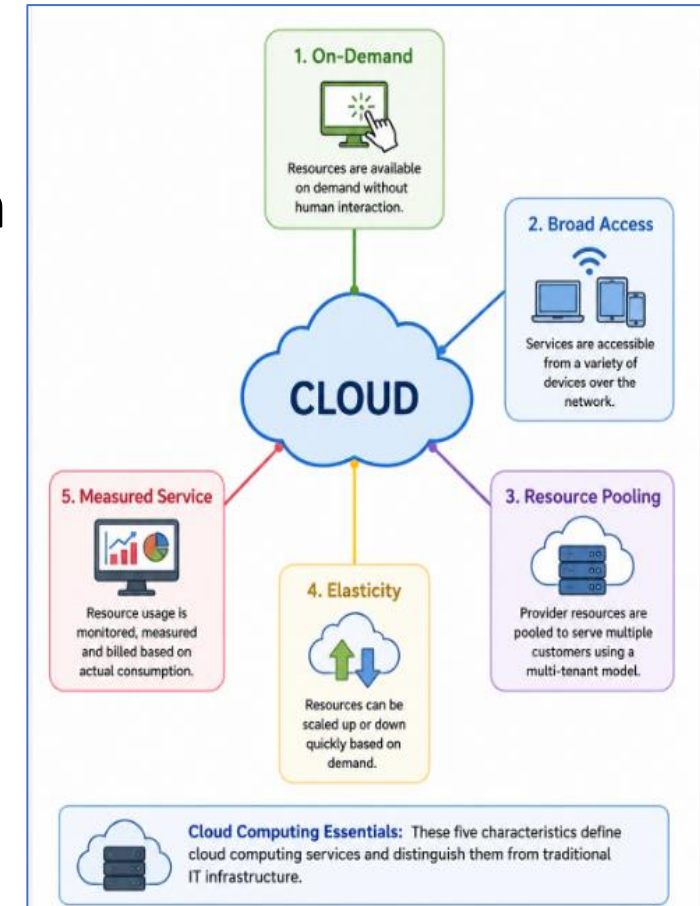
Key characteristics include:

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

CLOUD TECHNOLOGIES

Characteristics of Cloud Computing

- These features allow users to access computing resources whenever needed without direct interaction with the service provider.
- Cloud systems automatically allocate resources based on user demand while monitoring usage for billing purposes.
- **Five NIST characteristics:** On-Demand Self-Service, Broad Network Access, Resource Pooling, Rapid Elasticity, Measured Service



CLOUD TECHNOLOGIES

Applications of Cloud Computing

- Cloud computing is widely used in various sectors.

Examples include:

- Education (Google Classroom, Microsoft Teams)
- Banking (Online Banking)
- Healthcare (Electronic Health Records)
- E-commerce (Amazon)
- Entertainment (Netflix, Spotify)

CLOUD TECHNOLOGIES

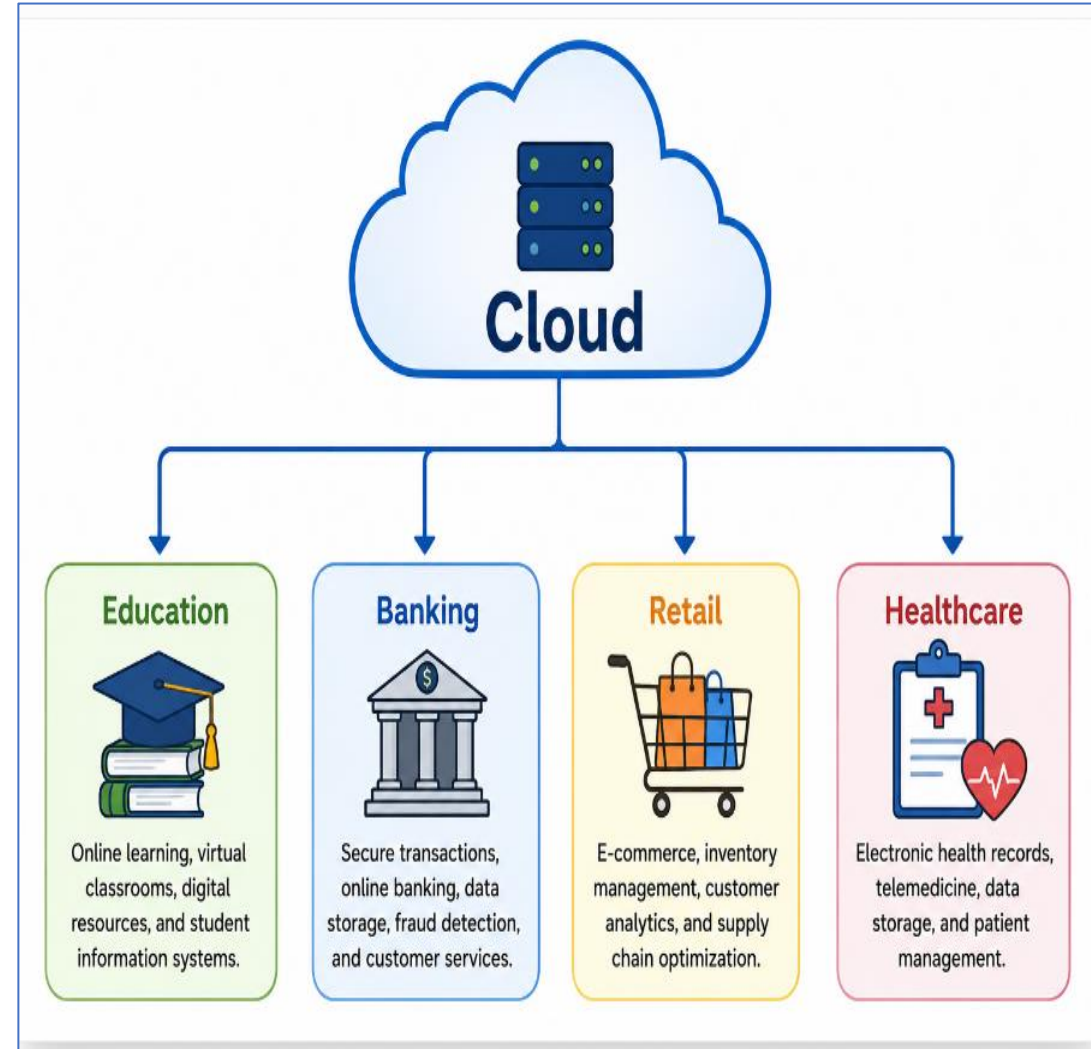
Applications of Cloud Computing

- Social Media (Facebook, Instagram)
- Software Development
- Artificial Intelligence
- Cloud computing enables organizations to provide reliable, secure, and scalable services to millions of users worldwide.
- Measured service

CLOUD TECHNOLOGIES

Applications of Cloud Computing

- Imagine the cloud as a water tank. Many buildings receive water from the same tank, but each uses it for different purposes. Similarly, one cloud infrastructure provides computing services to education, banking, retail, healthcare, and many other sectors



CLOUD TECHNOLOGIES

Benefits of Cloud Computing

Benefit	Real-World Example
Cost Savings	A startup uses Amazon Web Services instead of buying expensive servers, reducing its initial investment.
High Speed	A developer creates a virtual server on Microsoft Azure in a few minutes instead of waiting days to install physical hardware.
Scalability	An online shopping website increases its server capacity during festival sales and reduces it after the sale ends.
Anywhere Access	Students can access their assignments stored in Google Drive from a laptop, tablet, or mobile phone anywhere with an Internet connection.
Improved Security	Banks use cloud security features such as encryption and multi-factor authentication to protect customer data.
Easy Collaboration	Multiple students work together on the same document using Google Docs in real time.
Backup & Disaster Recovery	If a computer crashes, files stored in the cloud can be restored from the backup without losing important data.

CLOUD TECHNOLOGIES

Review

- **Activity**
- Students to identify five cloud-based applications installed on their smartphones.

CLOUD TECHNOLOGIES

Session Summary

- Definition of cloud computing.
- Need for cloud computing.
- Evolution of computing.
- Characteristics.
- Real-world applications.
- Benefits to users and organizations.

Review Questions

- Define cloud computing.
- Why has cloud computing become popular?
- Explain the evolution of computing.
- List five characteristics of cloud computing.
- Name five cloud applications.

- **Quick Quiz(True/False)**
- Cloud computing requires users to own physical servers. (False)
- Google Drive is a cloud application. (True)
- Cloud resources are accessible through the Internet. (True)
- Cloud computing is less scalable than traditional computing. (False)



THANK YOU

Shivaleela B H
Computer Applications



CLOUD TECHNOLOGIES

Shivaleela B H
Computer Applications



CLOUD COMPUTING

Traditional vs. Cloud Computing & History of Cloud Computing

Shivaleela B H
Computer Applications

Traditional vs. Cloud Computing & History of Cloud Computing

At the end of this session, students will be able to:

- Explain the concept of traditional computing.
- Compare traditional computing with cloud computing.
- Understand the limitations of traditional IT infrastructure.
- Describe the historical evolution of cloud computing.
- Identify major milestones in cloud computing.
- Recognize the contribution of virtualization and the Internet to cloud computing.

CLOUD TECHNOLOGIES

Traditional vs. Cloud Computing & History of Cloud Computing

- **Traditional computing** served organizations for many years but had limited flexibility and scalability.
- **Cloud computing** emerged to meet the growing demand for Internet-based and on-demand computing resources.

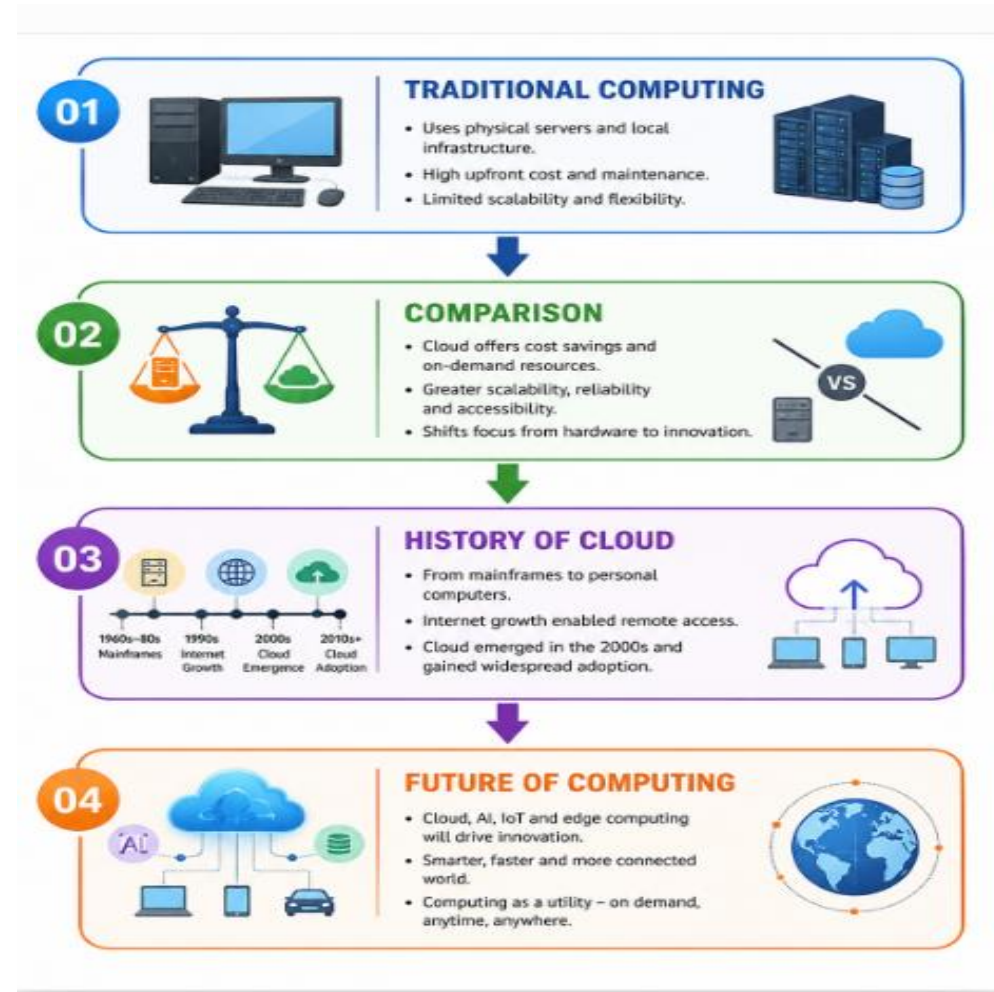
“Understanding the past helps us appreciate current technologies”

CLOUD TECHNOLOGIES

Traditional vs. Cloud Computing & History of Cloud Computing

Key Points

- Define traditional computing
- Compare traditional and cloud models
- Learn the history of cloud computing
- Understand major milestones
- Explore modern cloud platforms



CLOUD TECHNOLOGIES

Traditional Computing

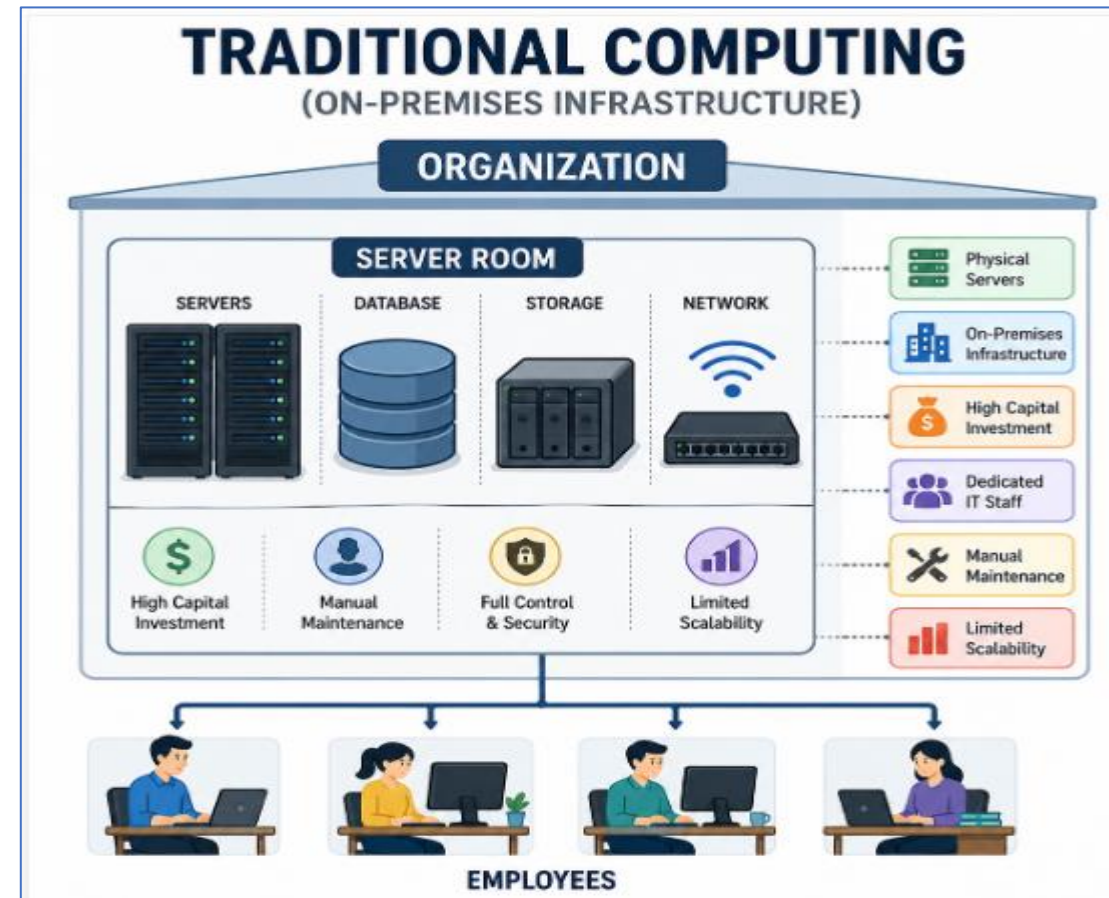
- **Traditional computing** uses physical servers, computers, storage, and networking devices installed within an organization's premises (on-premises).
- **Organizations own and manage** all hardware and software, including installation, maintenance, upgrades, and security.
- It requires a **high initial investment** in infrastructure and skilled IT professionals for system management.

CLOUD TECHNOLOGIES

Traditional Computing

Traditional Computing Characteristics

- Physical servers
- On-premises infrastructure
- High capital investment
- Dedicated IT staff
- Manual maintenance
- Limited scalability



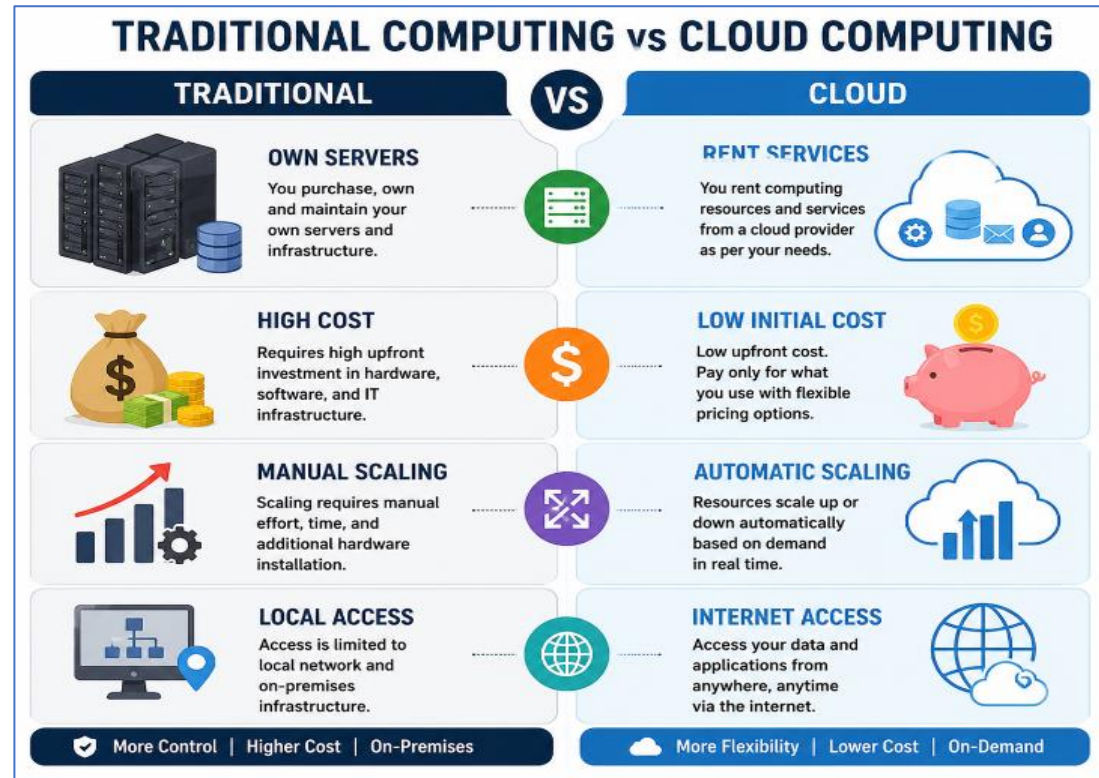
A university maintaining its own data centre and computer laboratories.

CLOUD TECHNOLOGIES

Traditional Computing vs. Cloud Computing

- Cloud computing enables organizations to focus on innovation rather than infrastructure management

Feature	Traditional Computing	Cloud Computing
Infrastructure	Owned	Rented
Cost	High Capital Cost	Pay-as-you-use
Maintenance	Customer	Cloud Provider
Scalability	Limited	Highly Scalable
Accessibility	Mostly Local Network	Internet
Deployment	Weeks/Months	Minutes
Backup	Manual	Automated
Availability	Limited	High Availability



Which model would be more suitable for a startup company and why?

CLOUD TECHNOLOGIES

Limitations of Traditional Computing

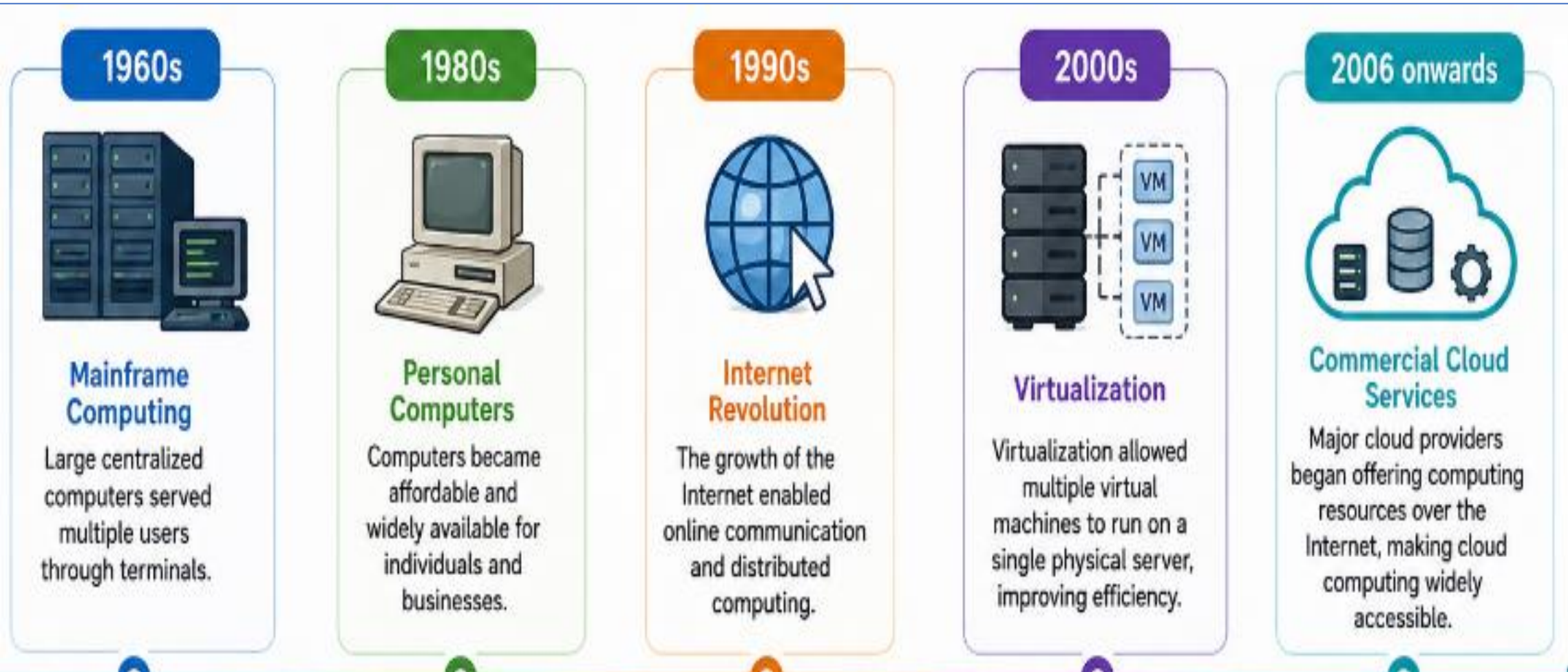
- Although traditional computing provides full control over hardware and software, it presents several challenges for modern businesses.
- Major limitations include:
- High infrastructure costs
- Long deployment time
- Hardware maintenance
- Limited scalability
- High electricity consumption

- Need for dedicated IT staff
- Complex disaster recovery
- Software licensing expenses
- As businesses grow, expanding traditional infrastructure requires purchasing additional servers and networking equipment, increasing operational costs and complexity.
- Cloud computing overcomes many of these issues by offering flexible, scalable, and cost-effective services.

CLOUD TECHNOLOGIES

History of Cloud Computing

Evolution



- **Major Milestones in Cloud Computing**
- **1961:** John McCarthy proposed computing as a **public utility**.
- **1999:** Salesforce introduced **Software as a Service (SaaS)**.
- **2002:** Amazon launched its first **Web Services** for developers.
- **2006:** Amazon Web Services introduced **Elastic Compute Cloud (EC2)**, making cloud infrastructure widely available.
- **2008:** Google launched **App Engine** for building cloud applications.

CLOUD TECHNOLOGIES

Major Milestones

- **2010:** Microsoft launched **Azure**, expanding enterprise cloud computing.
- **Present: AI, IoT, Edge Computing, and Serverless Computing** continue to drive cloud innovation..
- These milestones transformed cloud computing into the backbone of modern digital services.

CLOUD TECHNOLOGIES

Summary

In this session, we explored:

- Traditional computing
- Characteristics of on-premises infrastructure
- Traditional vs. cloud computing
- Limitations of traditional systems
- History of cloud computing & Major milestones
- Real-world cloud adoption through the Netflix case study

CLOUD TECHNOLOGIES

Review

- Define traditional computing.
- Differentiate between traditional computing and cloud computing.
- List four limitations of traditional computing.
- Explain the role of virtualization in cloud computing.
- Describe three major milestones in the history of cloud computing.
- Why did Netflix migrate to the cloud?

- Choose the correct answer:
- Which model allows users to rent computing resources?
 - A. Traditional Computing
 - B. Cloud Computing
 - C. Standalone Computing
 - D. Mainframe Computing

- virtual machines on one server?
 - A. Networking
 - B. Virtualization
 - C. Database
 - D. Compiler

- Which cloud provider launched EC2 in 2006?
 - A. Microsoft
 - B. Amazon Web Services
 - C. IBM
 - D. Oracle



THANK YOU

Shivaleela B H
Computer Applications



CLOUD TECHNOLOGIES

Shivaleela B H
Computer Applications



CLOUD COMPUTING

Cloud Computing Ecosystem

Shivaleela B H
Computer Applications

CLOUD TECHNOLOGIES

Cloud Computing Ecosystem

At the end of this session, students will be able to:

- Define the cloud computing ecosystem.
- Identify the participants in the ecosystem.
- Explain the responsibilities of each participant.
- Describe how cloud services are delivered.
- Analyze a real-world cloud ecosystem.

CLOUD TECHNOLOGIES

Cloud Computing Ecosystem

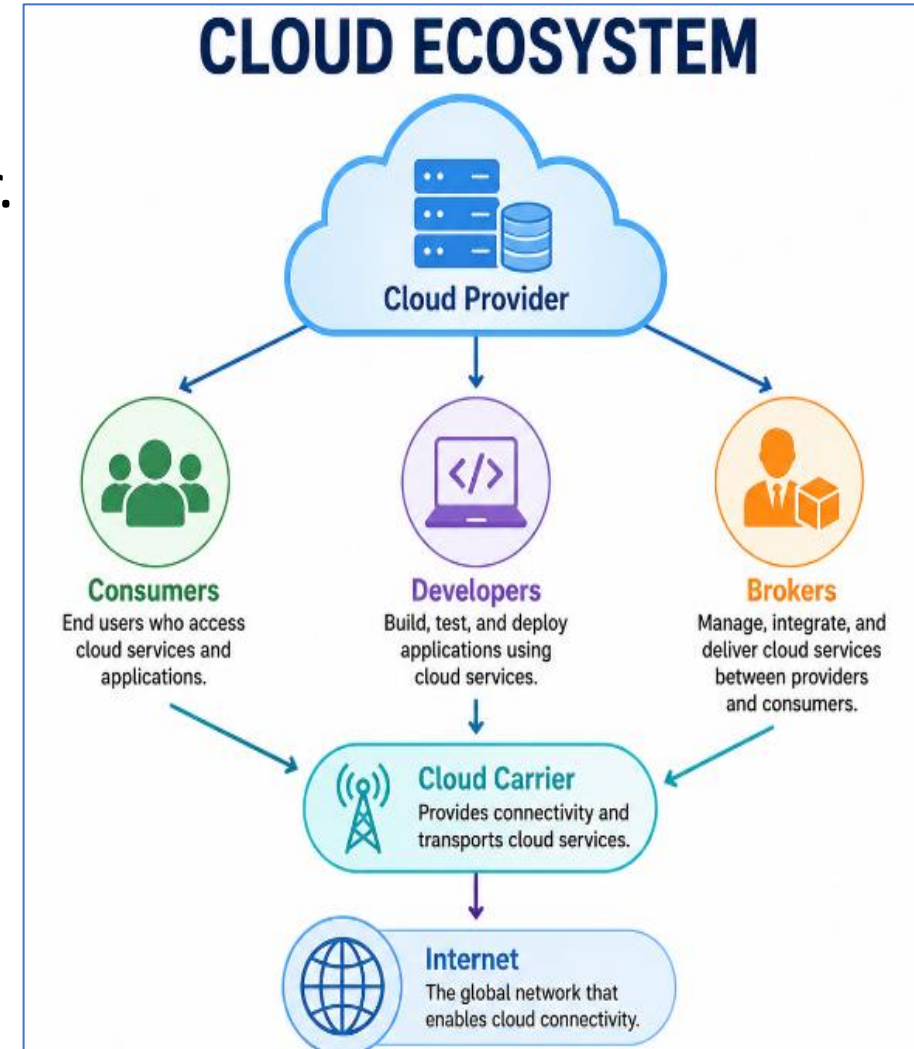
Key Points

- Definition of cloud ecosystem
- Main participants
- Roles and responsibilities
- Service delivery process
- Real-world examples

Cloud computing is much more than servers.

It involves several participants working together.

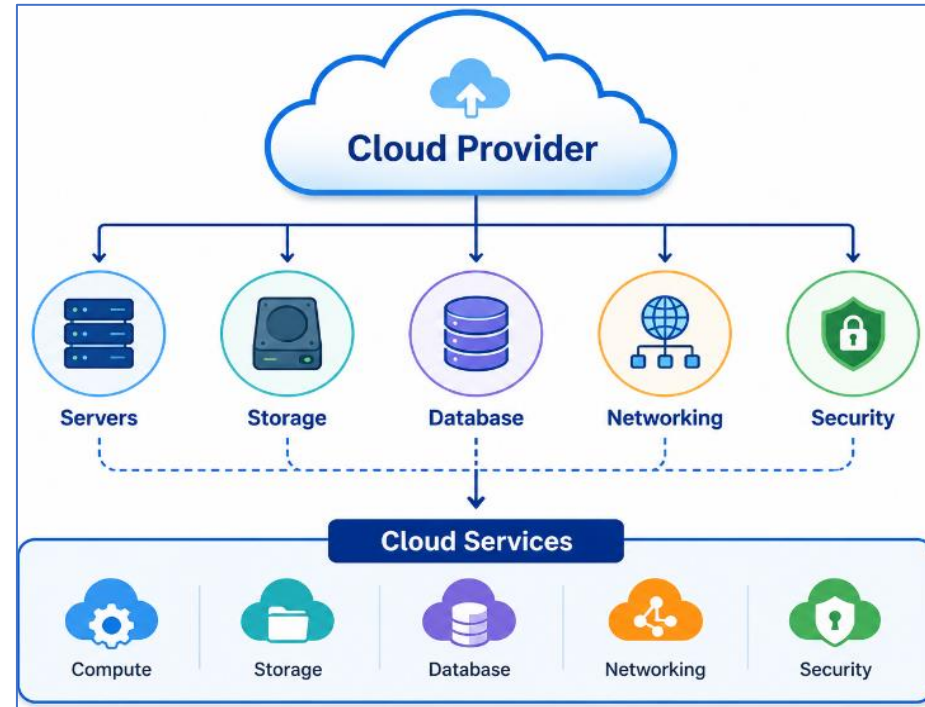
It is like online shopping platform where customers, sellers, delivery partners, payment gateways, and warehouses all contribute to the final service.



CLOUD TECHNOLOGIES

Main Participants: CSP

Cloud Service Providers: (CSP) owns and manages cloud infrastructure and delivers services over the Internet. These providers offer Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).



CLOUD TECHNOLOGIES

Main Participants: CSP

Cloud Provider Responsibilities	Leading Cloud Providers
Manage data centres	Amazon Web Services (AWS)
Maintain servers	Microsoft Azure
Provide storage services	Google Cloud
Offer database services	IBM Cloud
Ensure security and compliance	Oracle Cloud Infrastructure
Monitor service availability	
Generate customer billing	
Provide technical support	

Cloud Consumers: Individuals, businesses, educational institutions, or government organizations that use cloud services. Consumers access cloud resources through:

- Web browsers
- Mobile applications
- APIs
- Desktop software

Consumers pay only for the services they use.

Responsibility	Description	Example
Select the appropriate cloud service	Choose the right cloud model (IaaS, PaaS, or SaaS) based on business or personal needs.	A startup chooses AWS EC2 (IaaS) to host its web application.
Manage user accounts and access permissions	Create and control user accounts, roles, and permissions to ensure secure access.	A company uses Microsoft 365 to assign employee email accounts and access rights.
Protect sensitive data	Secure data using encryption, backups, and other security measures.	A hospital encrypts patient records and stores backups in the cloud.
Monitor service usage and costs	Track resource usage and spending to optimize performance and avoid unnecessary expenses.	A business monitors its monthly AWS bill and scales down unused virtual machines.
Comply with legal and regulatory requirements	Ensure cloud usage follows industry regulations and organizational policies.	A bank stores customer data in compliance with financial regulations using secure cloud services.

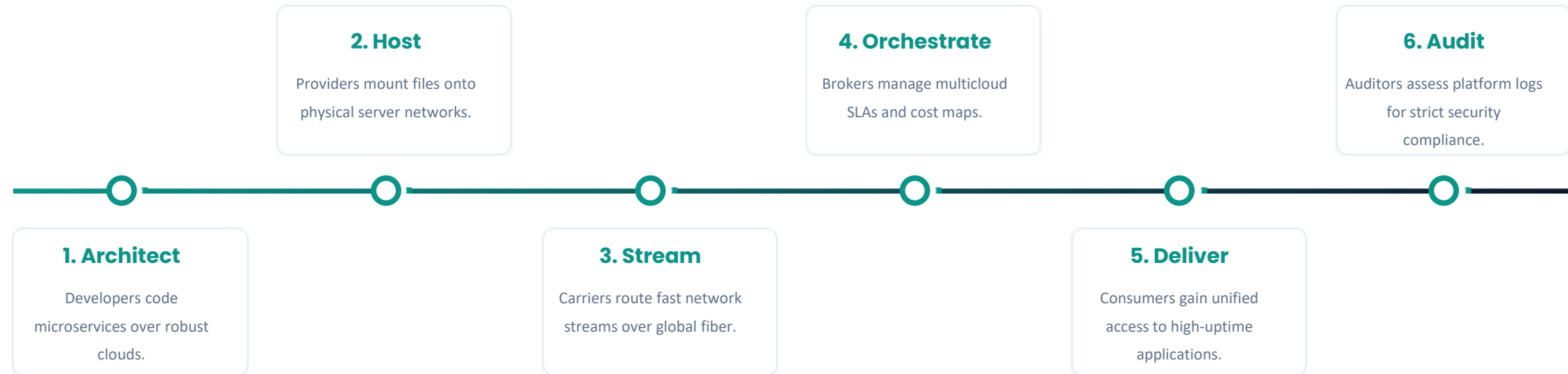
Supporting Roles

Ecosystem Participant	Core Operational Functions & Responsibilities	Food App Analogy
Cloud Developer	Architects, codes, and deploys scalable user applications.	The Kitchen Chef
Cloud Broker	Optimizes resource costs, integrates services, and handles SLAs.	The Food Ordering Platform
Cloud Auditor	Independent evaluation of platform security standards and compliance.	The Health & Safety Inspector
Cloud Carrier	Deploys raw network connectivity and packet routing (ISPs).	The Delivery Driver

CLOUD TECHNOLOGIES

Ecosystem Architecture Lifecycle

The cloud ecosystem follows a layered architecture where different participants communicate through the Internet.



Google Workspace

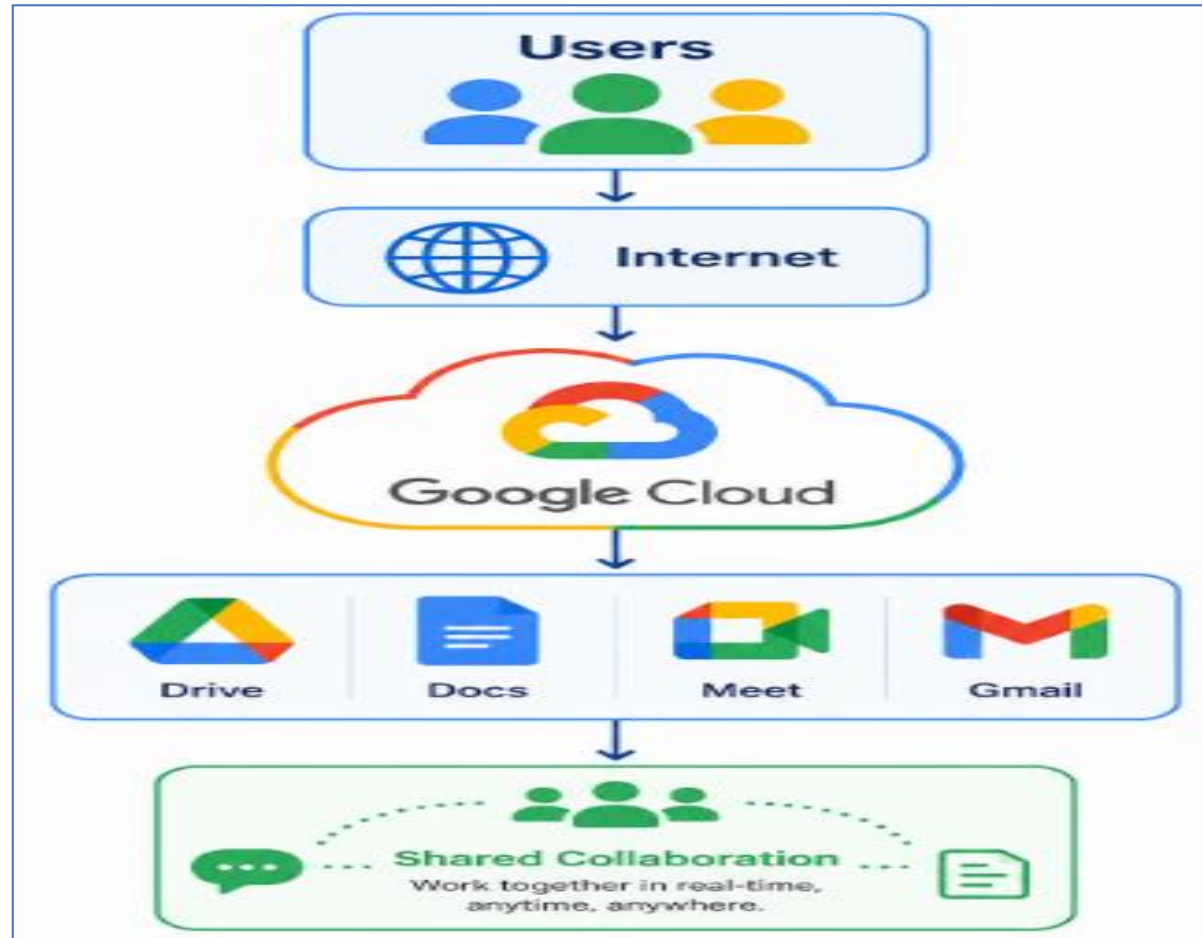
Ex: Google Cloud is a multi-tier cloud ecosystem where **Google Cloud Platform** acts as the **provider** by hosting computing and identity services. **Consumers** include millions of schools, students, and businesses who use applications such as **Gmail, Google Drive, Google Docs, Calendar, and Google Meet.**

These services are delivered through **Internet Service Providers (ISPs)**, which act as carriers, while **third-party auditors** ensure compliance with security standards such as **SOC 2** and **HIPAA.**

CLOUD TECHNOLOGIES

Case Study

Google Workspace



Session Summary

- ✓ Cloud ecosystems integrate hardware, software, and people to deliver cloud services.
- ✓ Cloud Service Providers (CSPs) manage data centers and cloud infrastructure.
- ✓ Brokers, Carriers, and Auditors support cloud service delivery and security.
- ✓ All ecosystem components work together to provide reliable and scalable services.

Review Questions

1. Define the cloud computing ecosystem in your own words.
2. Differentiate between a Cloud Broker and a Cloud Carrier.
3. What is the role of a Cloud Auditor?
4. Explain the Google Workspace ecosystem with an example.

Which participant provides Internet connectivity? A. Developer B. Cloud Carrier C. Broker D. Auditor	Who develops cloud applications? A. Consumer B. Developer C. Carrier D. Auditor	Which participant checks compliance and security? A. Provider B. Consumer C. Auditor D. Carrier
---	--	--



THANK YOU

Shivaleela B H
Computer Applications



CLOUD TECHNOLOGIES

Shivaleela B H
Computer Applications

CLOUD COMPUTING

Characteristics of Cloud Computing

Shivaleela B H
Computer Applications

CLOUD TECHNOLOGIES

Characteristics of Cloud Computing

At the end of this session, students will be able to:

- Define the characteristics of cloud computing.
- Explain the five essential characteristics identified by NIST.
- Differentiate cloud computing from traditional computing based on these characteristics.
- Identify practical examples of each characteristic.
- Evaluate how these characteristics benefit organizations and users.

CLOUD TECHNOLOGIES

Characteristics of Cloud Computing

- Unique characteristics of cloud computing enable organizations to deliver IT services efficiently, reduce costs, improve flexibility, and meet changing business demands. NIST identifies five essential characteristics of cloud computing that distinguish it from traditional computing.

Key Points

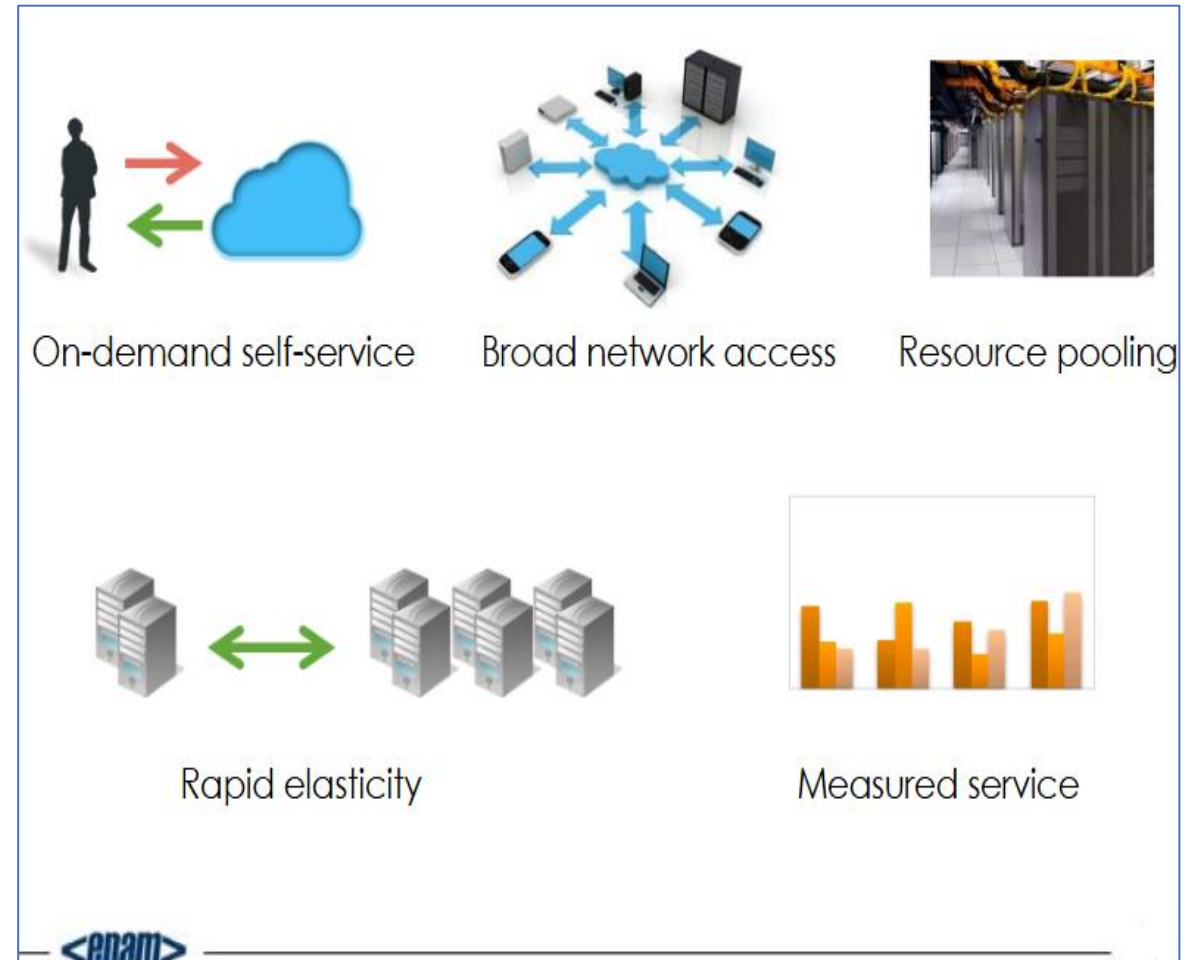
- NIST characteristics
- Flexible resource management
- Efficient service delivery
- Practical applications

CLOUD TECHNOLOGIES

Characteristics of Cloud Computing

NIST Core Characteristics Overview

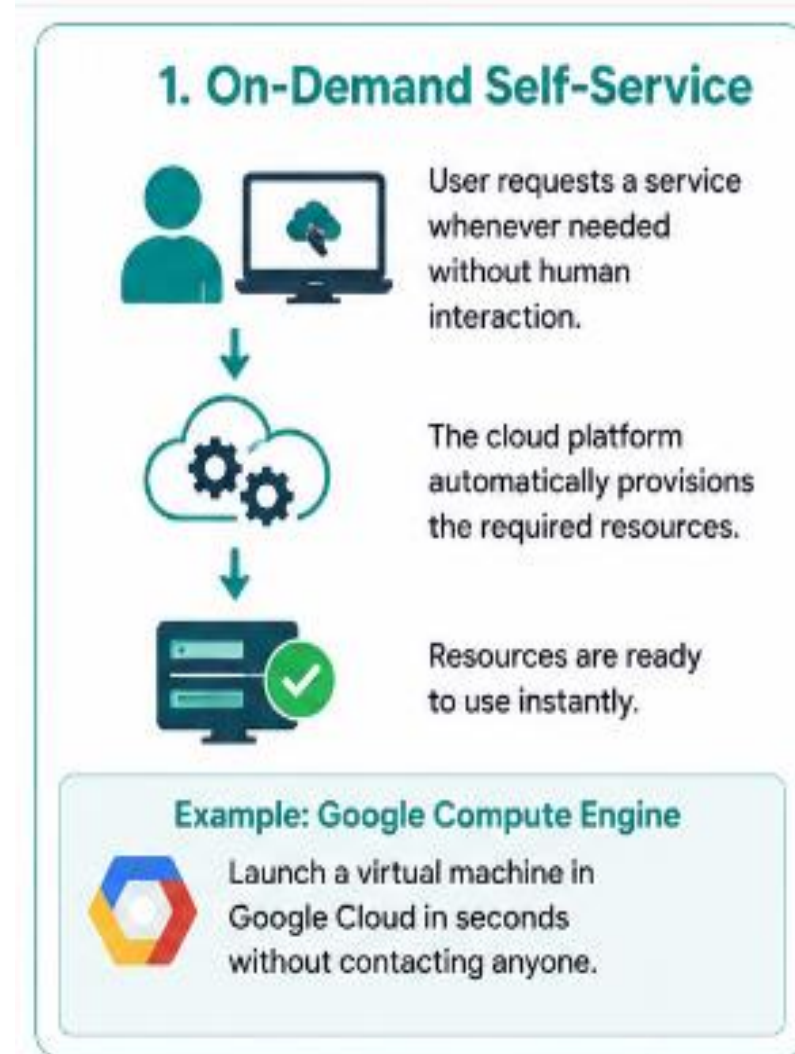
- On-Demand Self-Service
- Broad Network Access
- Resource Pooling
- Rapid Elasticity
- Measured Service



CLOUD TECHNOLOGIES

On-Demand Self-Service

- Users can provision computing resources automatically.
- No human intervention required from the service provider.
- Allows instant deployment of servers, storage, and software.



CLOUD TECHNOLOGIES

Broad Network Access

- Cloud capabilities are available over the network.
- Accessible through standard mechanisms and diverse devices.
- Supports smartphones, tablets, laptops, and workstation PCs.

2. Broad Network Access



Services are accessible over the network using standard protocols.



Access from a variety of devices (mobiles, laptops, desktops, tablets).



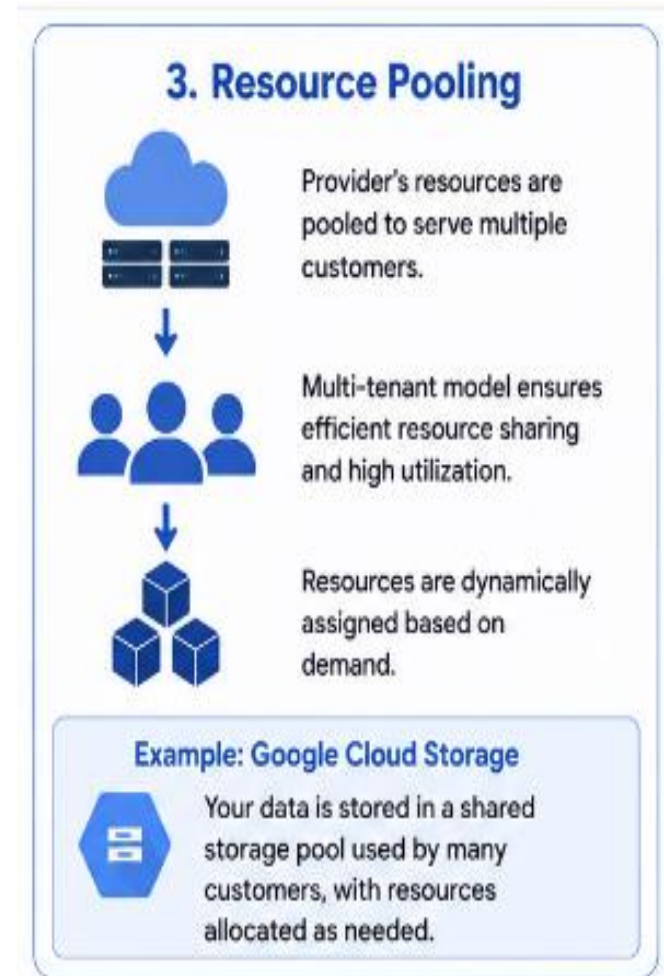
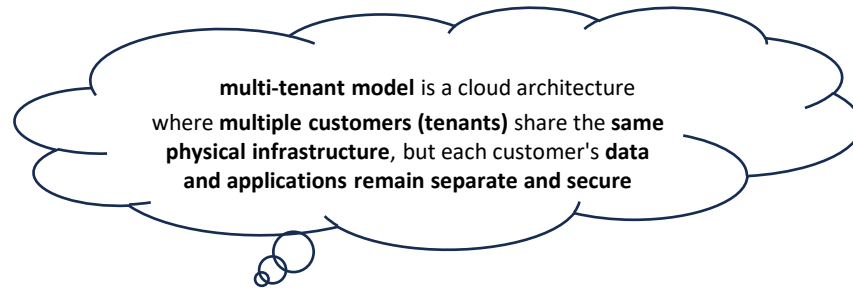
Consistent experience anytime, anywhere.

Example: Google Drive



Access your files on Google Drive from your laptop at home, phone on the go, or tablet at work.

- Provider resources are pooled to serve multiple consumers.
- Utilizes a highly efficient multi-tenant operational model.
- Physical and virtual resources dynamically assigned based on demand.



- Allows automatic scaling up or down instantly with demand peaks.
- Capabilities can be elastically provisioned and released.
- Appears seamless and unlimited to the end consumer.

4. Rapid Elasticity



Resource demand increases or decreases.



The cloud system rapidly scales resources up or down.



Resources match demand in near real-time.

Example: Google Kubernetes Engine (GKE)

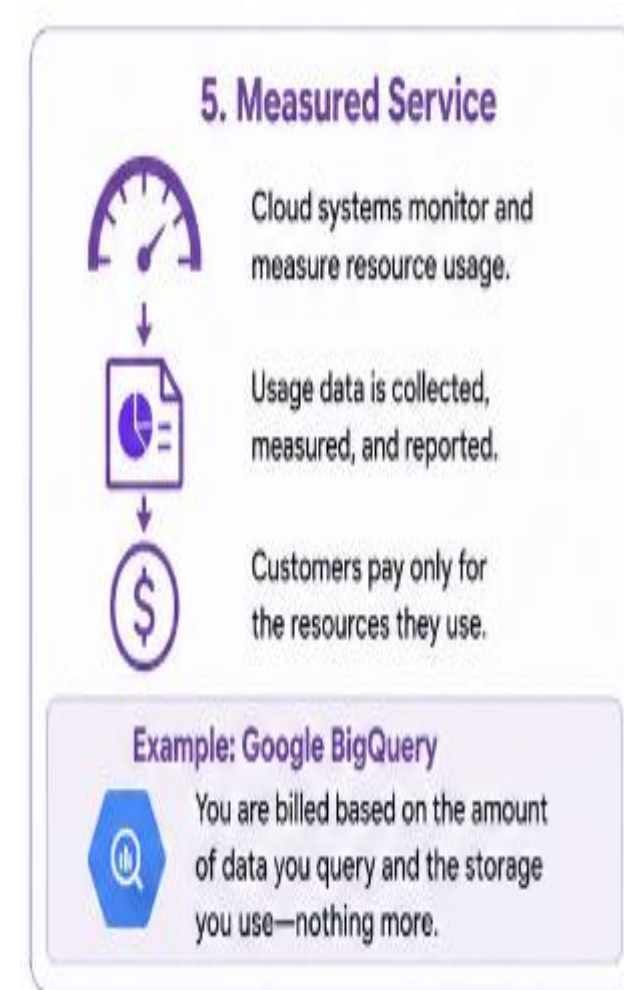


If your app gets more traffic, GKE automatically adds more containers (instances). When traffic drops, it removes them.

CLOUD TECHNOLOGIES

Measured Services

- Cloud systems automatically control and optimize resource usage.
- Leverages a metering capability tailored to the type of service.
- Provides full transparency through precise pay-as-you-go billing.



CLOUD TECHNOLOGIES

NIST Characteristics Summary

- The five essential characteristics work together to make cloud computing flexible, efficient, and scalable.

Characteristic	Meaning	Example
On-Demand Self-Service	Users provision resources themselves	Create a VM in Azure
Broad Network Access	Access via Internet	Google Drive
Resource Pooling	Shared infrastructure	Multi-tenant data center
Rapid Elasticity	Automatic scaling	Online shopping during sales
Measured Service	Pay for usage	AWS monthly billing

Session Summary

- ✓ Defined the characteristics of cloud computing.
- ✓ Learned the five NIST essential characteristics.
- ✓ Studied practical examples of each characteristic.
- ✓ Understood the importance of cloud characteristics for businesses and users.

Review Questions

1. What is On-Demand Self-Service?
2. Explain Broad Network Access with an example.
3. What is Resource Pooling?
4. Differentiate Scalability and Rapid Elasticity.
5. Why is Measured Service important?
6. List the five NIST characteristics of cloud computing.

Question 1	Question 2	Question 3
Which characteristic allows users to create resources without contacting the provider? ○ A. Resource Pooling ○ B. On-Demand Self-Service ○ C. Measured Service ○ D. Broad Network Access	Which characteristic enables automatic scaling based on demand? ○ A. Measured Service ○ B. Rapid Elasticity ○ C. Resource Pooling ○ D. Broad Network Access	Which characteristic supports pay-as-you-use billing? ○ A. Resource Pooling ○ B. Broad Network Access ○ C. Measured Service ○ D. On-Demand Self-Service



THANK YOU

Shivaleela B H
Computer Applications



CLOUD TECHNOLOGIES

Shivaleela B H
Computer Applications



CLOUD COMPUTING

Advantages & Disadvantages of Cloud Computing

Shivaleela B H
Computer Applications

CLOUD TECHNOLOGIES

Advantages & Disadvantages of Cloud Computing

At the end of this session, students will be able to:

- Advantages of cloud computing.
- Explain the Describe the disadvantages and risks of cloud computing.
- Analyze the impact of cloud adoption on businesses.
- Compare the benefits and challenges.
- Recommend appropriate cloud solutions for different scenarios.

CLOUD TECHNOLOGIES

Advantages & Disadvantages of Cloud Computing

- Cloud computing offers significant advantages such as cost savings, scalability, and flexibility, it also presents challenges related to security, privacy, Internet dependence, and compliance.
- Understanding both the benefits and limitations helps organizations make informed decisions about cloud adoption.

Key Points

- Benefits of cloud computing
- Challenges and risks
- Business impact
- Decision-making

CLOUD TECHNOLOGIES

Advantages of Cloud Computing

- Cloud computing provides numerous advantages that make it attractive for businesses, educational institutions, governments, and individuals.
- **Major Advantages**
 - Cost Savings
 - No need to purchase expensive hardware.
 - Reduced maintenance costs.
 - Pay only for the resources used.

CLOUD TECHNOLOGIES

Advantages of Cloud Computing

- **Scalability:** Increase or decrease resources according to demand.
- **Accessibility:** Access applications and data from anywhere using the Internet.
- **Automatic Updates:** Cloud providers regularly update software and security patches.
- **High Availability:** Services are designed for continuous operation with minimal downtime.
- **Collaboration:** Multiple users can work on the same document simultaneously.

CLOUD TECHNOLOGIES

Benefits of Cloud Computing

- Cloud computing supports organizations by improving efficiency, reducing costs, and enabling innovation.
- **Business Benefits**
 - Faster deployment of applications.
 - Improved disaster recovery.
 - Enhanced data backup.
 - Better business continuity.
 - Global accessibility.
 - Reduced IT infrastructure management.
 - Support for remote work and Increased productivity.

- Organizations can launch new services quickly without investing in physical servers. During periods of high demand, cloud resources can be automatically scaled to maintain performance.
- **Business Scenario**
 - A retail company experiences increased website traffic during festival sales. Cloud computing automatically increases server capacity to handle the additional visitors and scales down once the sale ends.

- Although cloud computing offers many benefits, organizations must also consider its limitations.
- **Common Disadvantages:** Internet Dependency , Cloud services require a reliable Internet connection.
- **Security Risks :**Data stored on remote servers may be vulnerable to cyberattacks if not properly protected.
- **Privacy Concerns :**Sensitive information may be stored in different countries.
- **Limited Control :**Customers have less control over hardware and infrastructure.

CLOUD TECHNOLOGIES

Disadvantages of Cloud Computing

- **Downtime** :Service outages may temporarily interrupt access.
- **Vendor Lock-in** :Migrating from one cloud provider to another can be difficult and expensive.
- **Ex:** An online examination system becomes inaccessible due to an Internet outage.
- Organizations should have backup plans and disaster recovery strategies.

CLOUD TECHNOLOGIES

Security and Privacy Challenges

- Security is one of the most important considerations in cloud computing.
- **Common Security Threats**
 - Unauthorized access , Data breaches
 - Malware attacks, Phishing attacks
 - Insider threats, Weak passwords
 - Misconfigured cloud services

CLOUD TECHNOLOGIES

Security and Privacy Challenges

- **Privacy Challenges**
 - Data stored across multiple countries, Compliance with legal regulations.
 - Protection of personal information, Data ownership concerns.
- **Security Best Practices**
 - Multi-factor authentication (MFA) , Strong passwords , Data encryption
 - Regular backups ,Access control policies , Security monitoring

CLOUD TECHNOLOGIES

Advantages VS Disadvantages

Feature	Advantages	Disadvantages
Cost	Lower initial investment	Ongoing subscription costs
Accessibility	Access from anywhere	Depends on Internet
Maintenance	Managed by provider	Less infrastructure control
Scalability	Automatic resource scaling	Potential vendor lock-in
Collaboration	Easy real-time collaboration	Security considerations
Reliability	High availability	Possible service outages

CLOUD TECHNOLOGIES

Case Study

GOOGLE CLOUD CASE STUDY

Transforming Businesses. Solving Real-World Problems.

Google Cloud helps organizations innovate faster, operate more securely, and deliver better customer experiences using scalable, intelligent cloud solutions.



- Scalability**
Handle massive user growth and streaming demand.
- Data Management**
Process and analyze large volumes of data in real time.
- Reliability & Performance**
Ensure high availability and seamless user experience globally.
- Cost Optimization**
Reduce infrastructure costs while maintaining performance.

ABOUT THE CUSTOMER

GOOGLE CLOUD SOLUTION

SOLUTION ARCHITECTURE (OVERVIEW)



Industry: Media & Entertainment
Headquarters: Stockholm, Sweden

Spotify is a global music streaming platform with millions of users and a massive volume of data generated every day.

- Compute Engine**
Flexible virtual machines to support Spotify's workloads.
- Cloud Storage**
Durable, scalable storage for petabytes of data.
- BigQuery**
Real-time analytics on large datasets for better insights.
- Kubernetes Engine (GKE)**
Containerized application deployment for scalability and portability.
- AI & Machine Learning**
Personalized recommendations using ML models on Vertex AI.



The diagram shows the flow from Users, Apps & Services, and API APIs into the Google Cloud Platform. The platform includes Load Balancing, Compute Engine, Kubernetes Engine, Cloud Storage, BigQuery, and AI / ML (Vertex AI). These are supported by Security & Identity and Monitoring & Logging, all running on Global Infrastructure.

BUSINESS IMPACT

- Scalability Achieved**
Supports millions of users and thousands of new playlists daily.
- Faster Insights**
Real-time analytics helps Spotify make data-driven decisions.
- High Reliability**
99.9%+ uptime ensures seamless music streaming worldwide.
- Cost Efficiency**
Optimized infrastructure reduced costs significantly.

KEY RESULTS

- 500M+**
Active users supported with high performance
- Real-time**
Data analytics for better recommendations
- 30%+**
Reduction in infrastructure costs
- Global Scale**
Reliable service across the world

WHY GOOGLE CLOUD?

- Global, secure, and reliable infrastructure
- Advanced data analytics and AI capabilities
- Scalable and flexible services
- Strong support for innovation and business growth

CONCLUSION

By leveraging Google Cloud, Spotify delivers a seamless music experience to millions of users worldwide, while continuously innovating and making data-driven improvements.

“Google Cloud empowers us to focus on innovation and our customers, while it handles the scale and complexity.”

– Spotify Engineering Team

Session Summary

- ✓ Defined the characteristics of cloud computing.
- ✓ Advantages of cloud computing.
- ✓ Business benefits.
- ✓ Cost savings and scalability.
- ✓ Disadvantages and challenges.
- ✓ Security and privacy issues.
- ✓ Comparison of advantages and disadvantages.

Review Questions

1. List five advantages of cloud computing.
2. Explain three disadvantages of cloud computing.
3. What is vendor lock-in?
4. Why is Internet connectivity important for cloud services?
5. How does cloud computing improve business continuity?
6. Explain the importance of data encryption in cloud computing.

Question 1	Question 2	Question 3
Which is a major advantage of cloud computing? A. Higher hardware costs B. Scalability C. Manual software updates D. Limited accessibility	Which is a cloud security best practice? A. Weak passwords B. Multi-factor authentication C. Sharing credentials D. Disabling backups	Which challenge refers to difficulty in changing cloud providers? A. Scalability B. Collaboration C. Vendor Lock-in D. Elasticity



THANK YOU

Shivaleela B H
Computer Applications



CLOUD TECHNOLOGIES

Shivaleela B H
Computer Applications



CLOUD COMPUTING

Business drivers, Application of cloud

Shivaleela B H
Computer Applications

CLOUD TECHNOLOGIES

Business Drivers and Applications of Cloud Computing

At the end of this session, students will be able to:

- Define business drivers for cloud adoption.
- Explain why organizations migrate to the cloud.
- Identify cloud applications in different industries.
- Analyse real-world cloud adoption cases.
- Evaluate the impact of cloud computing on businesses.

- A business driver is a factor that motivates an organization to adopt a new technology or improve existing processes.
- Cloud computing supports business goals such as digital transformation, business continuity, scalability, and faster service delivery.

Key Points

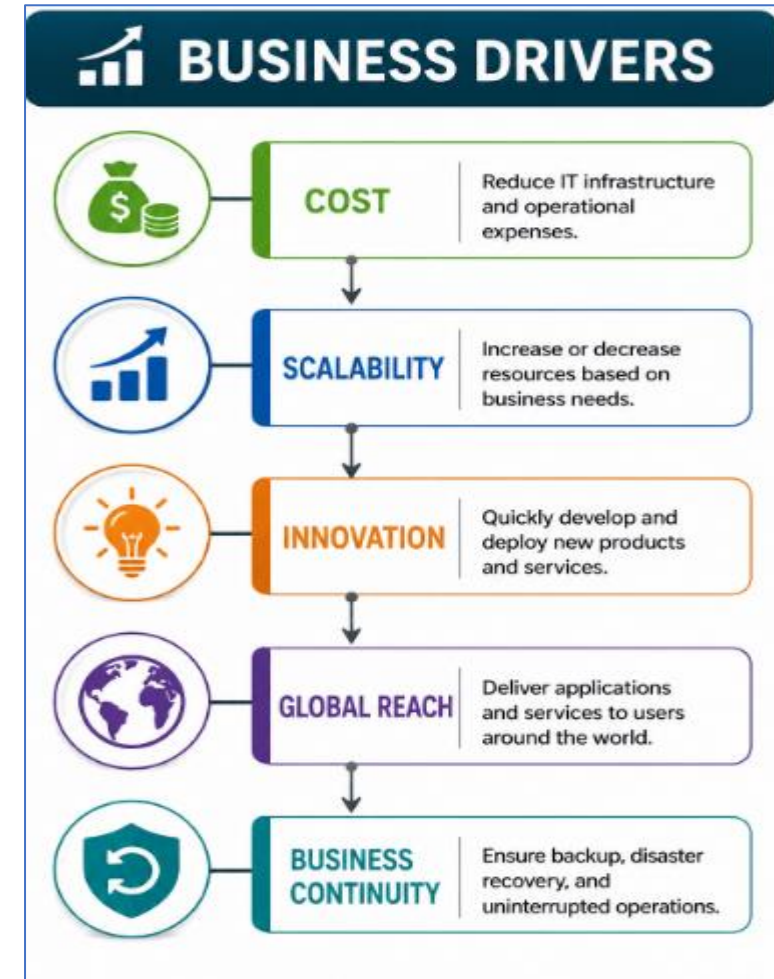
- Business drivers
- Digital transformation
- Industry applications
- Business value
- Real-world examples
-

CLOUD TECHNOLOGIES

Business Drivers of Cloud Computing

Organizations adopt cloud computing because it supports important business objectives.

- **Major Business Drivers**
- **Cost Optimization:** Lower investment in hardware and maintenance, Pay only for the resources used.
- **Scalability:** Easily increase or decrease resources according to demand.



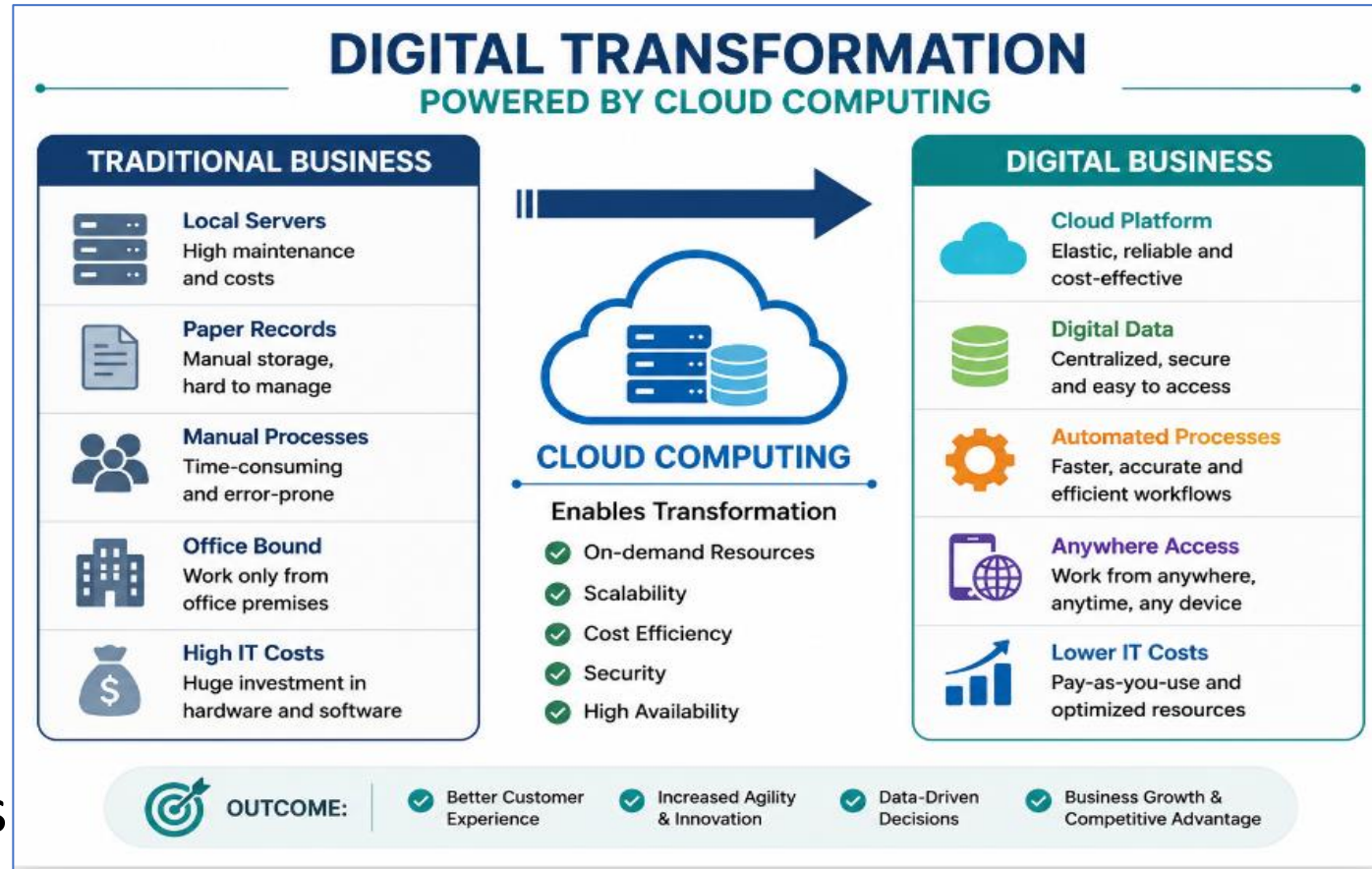
CLOUD TECHNOLOGIES

Business Drivers of Cloud Computing

- **Innovation:** Quickly experiment with new ideas using cloud platforms.
- **Business Agility:** Launch products and services quickly.
- **Global Reach:** Serve customers worldwide through distributed data centres.
- **Business Continuity:** Maintain operations during hardware failures or disasters.
- **Benefits:** Faster deployment , Reduced operational costs Better customer experience, Improved competitiveness

- Digital transformation is the process of using digital technologies to improve business operations, customer experiences, and decision-making.
- Cloud computing is a key enabler of digital transformation because it provides flexible computing resources and supports modern technologies such as:
 - Artificial Intelligence (AI)
 - Machine Learning (ML)

- Internet of Things (IoT)
- Big Data Analytics
- Mobile Applications
- Cloud platforms allow organizations to innovate faster, analyse large volumes of data, and deliver personalized services.



- **Advantages**
 - Faster innovation
 - Data-driven decision making
 - Improved collaboration
 - Better customer experiences
 - Automation of business processes

CLOUD TECHNOLOGIES

Applications of Cloud Computing in Education

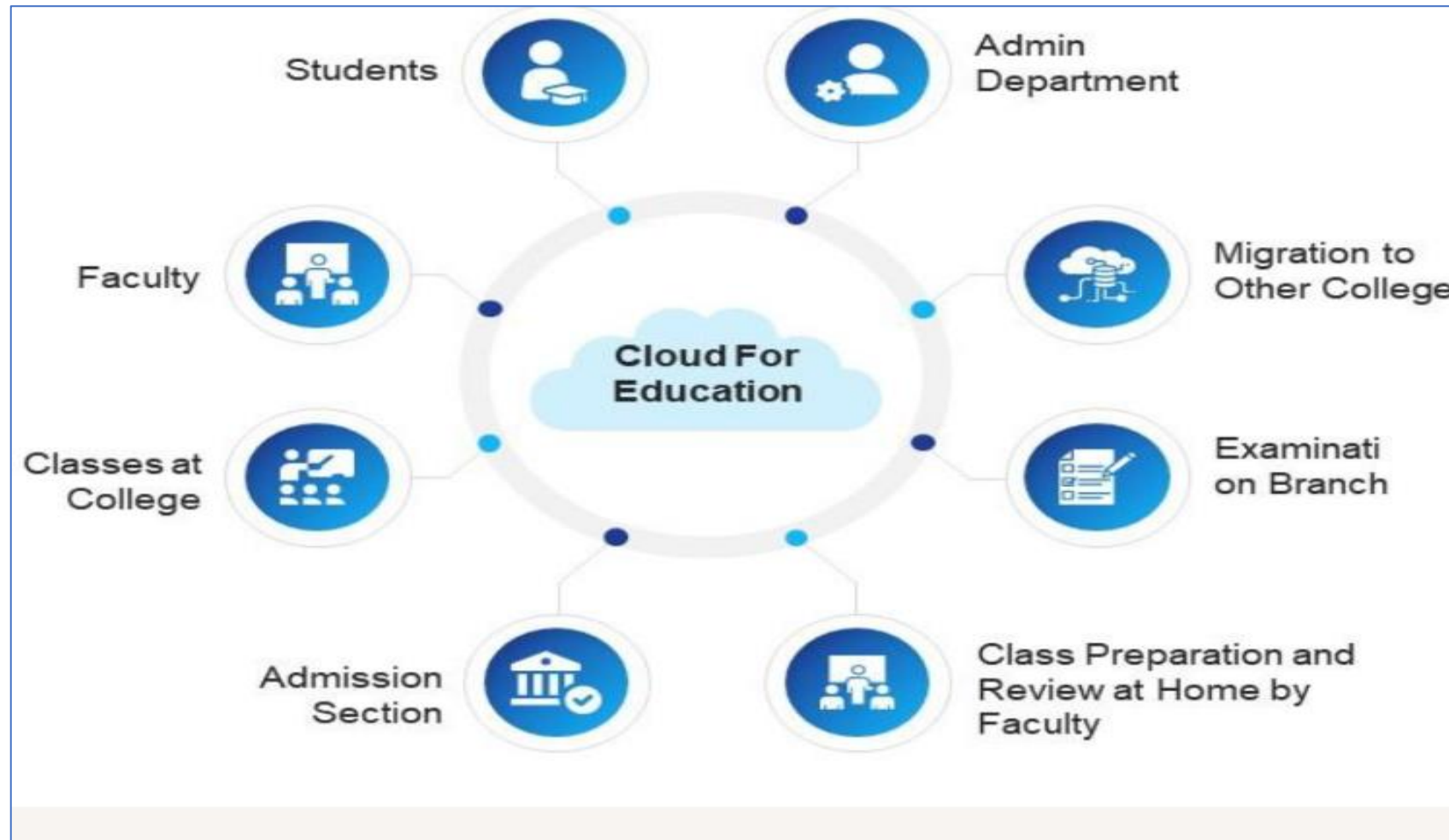
- Cloud computing has transformed education by enabling online learning, collaboration, and digital content delivery.

Category	Details
Applications	• Learning Management Systems (LMS) • Online Classrooms • Virtual Laboratories • Digital Libraries • Assignment Submission • Cloud Storage • Online Examinations • Student Information Systems
Benefits	• Access learning materials anytime, anywhere • Easy collaboration between students and teachers • Reduced paper usage • Secure storage of academic records • Supports hybrid and online learning
Popular Examples	Google Classroom, Microsoft Teams, Moodle, Zoom

CLOUD TECHNOLOGIES

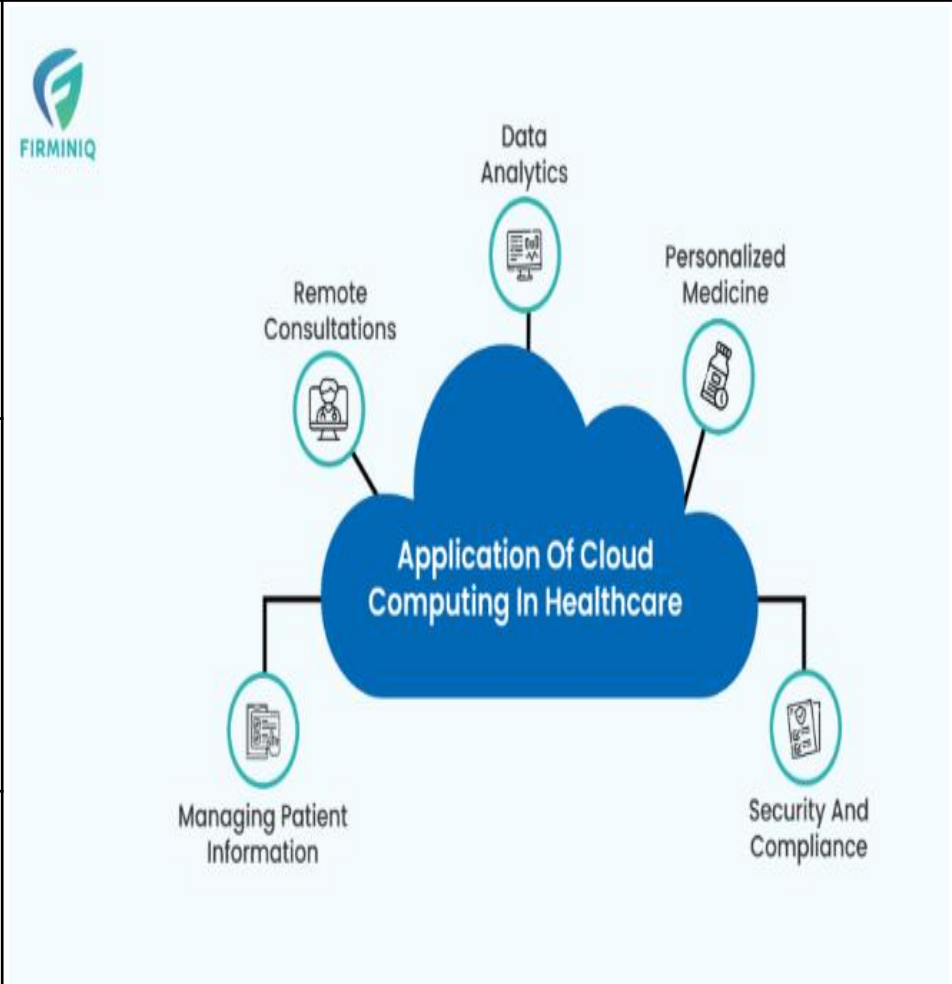
Applications of Cloud Computing in Education

- Cloud Based LMS



CLOUD TECHNOLOGIES

Applications in Healthcare

Healthcare	
Applications • Electronic Health Records (EHR) • Telemedicine • Medical Image Storage • Appointment Scheduling • Hospital Management Systems	
Benefits • Better patient care • Faster access to medical records • Secure storage of health data • Remote healthcare services	
Examples • Apollo 24/7 • Practo • e-Hospital (NIC)	

CLOUD TECHNOLOGIES

Applications in Banking

Banking	
Applications • Online Banking• Mobile Banking• Fraud Detection• Digital Payments• Customer Relationship Management (CRM)• Risk Analysis	
Benefits • Faster transactions• Improved customer service• Better data security• Disaster recovery• Anywhere banking access	
Examples • SBI YONO• HDFC NetBanking• Google Pay• PhonePe	

CLOUD TECHNOLOGIES

Netflix cloud adoption case study

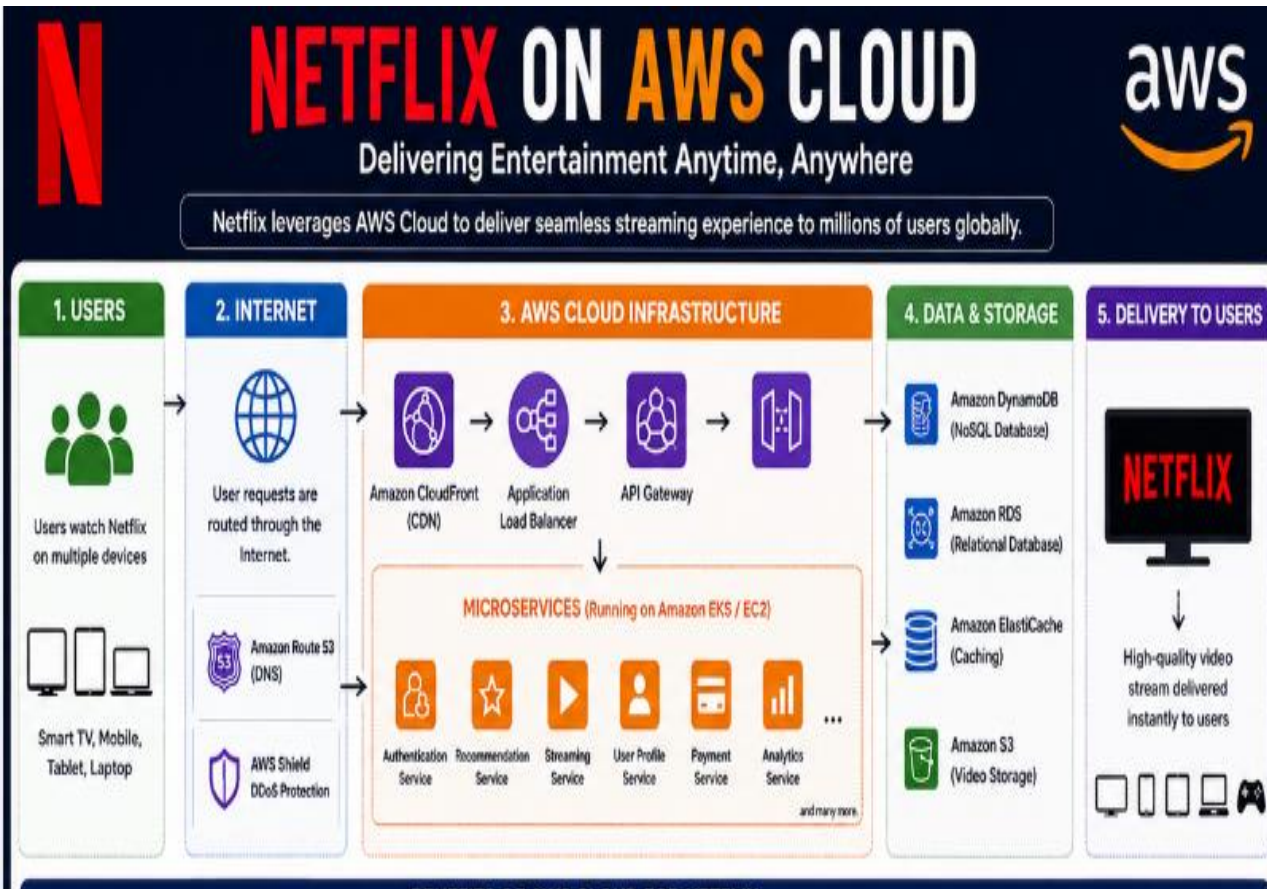
Netflix serves millions of users across different countries. To deliver uninterrupted streaming services, Netflix uses cloud infrastructure to manage large volumes of data and user requests

Challenges Before Cloud Adoption	Benefits After Cloud Adoption	Business Impact
• Increasing customer demand • High infrastructure costs • Limited scalability • Disaster recovery concerns	✓ Automatic resource scaling ✓ Global content delivery ✓ High availability ✓ Reduced infrastructure management ✓ Faster deployment of new features	➤ Better customer experience ➤ Lower operational costs ➤ Continuous innovation ➤ Reliable streaming services

CLOUD TECHNOLOGIES

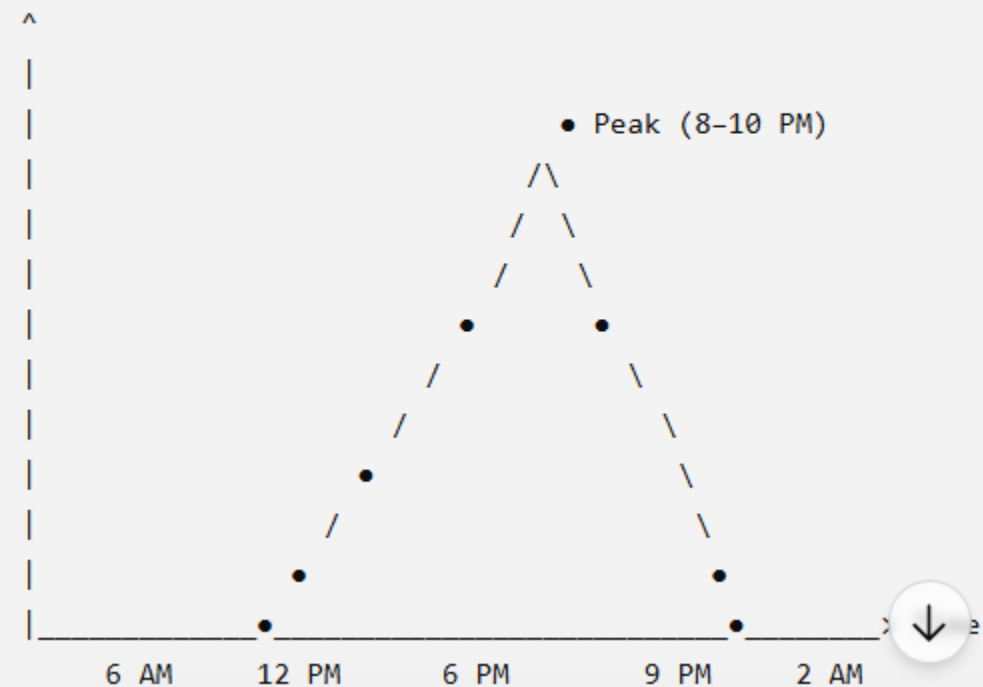
Netflix cloud adoption case study

Netflix experiences varying traffic throughout the day. Cloud computing allows automatic scaling during peak hours and cost savings during low-demand periods.



Netflix Traffic vs Time of Day

Users



Session Summary

- ✓ Business drivers for cloud adoption.
- ✓ Digital transformation.
- ✓ Cloud applications in education.
- ✓ Cloud applications in healthcare.
- ✓ Cloud applications in banking.
- ✓ Cloud applications in retail, entertainment, and government.
- ✓ Netflix cloud adoption case study.

Review Questions

1. What is a business driver?
2. List five business drivers for cloud adoption.
3. Explain digital transformation.
4. How does cloud computing benefit educational institutions?
5. Mention three applications of cloud computing in healthcare.
6. Why do banks use cloud computing?
7. Explain the benefits of cloud computing in retail.

Question 1	Question 2	Question 3
Which is a major business driver for cloud adoption? A. Increased hardware maintenance B. Cost optimization C. Manual scaling D. Limited accessibility	Which industry commonly uses Electronic Health Records (EHR)? A. Retail B. Healthcare C. Entertainment D. Manufacturing	Which cloud benefit allows businesses to quickly increase resources during high demand? A. Vendor Lock-in B. Scalability C. Downtime D. Manual Provisioning



THANK YOU

Shivaleela B H
Computer Applications



CLOUD TECHNOLOGIES

Shivaleela B H
Computer Applications



CLOUD COMPUTING

Cloud Deployment Models

Shivaleela B H
Computer Applications

CLOUD TECHNOLOGIES

Cloud Deployment Models

At the end of this session, students will be able to:

- Define cloud deployment models.
- Explain the four deployment models.
- Compare their advantages and disadvantages.
- Identify suitable business applications.
- Recommend the appropriate deployment model for different scenarios.

CLOUD TECHNOLOGIES

Cloud Deployment Models

- Cloud deployment models describe how cloud infrastructure is deployed, managed, owned, and accessed. Organizations choose a deployment model based on factors such as security, cost, scalability, compliance, and business requirements.

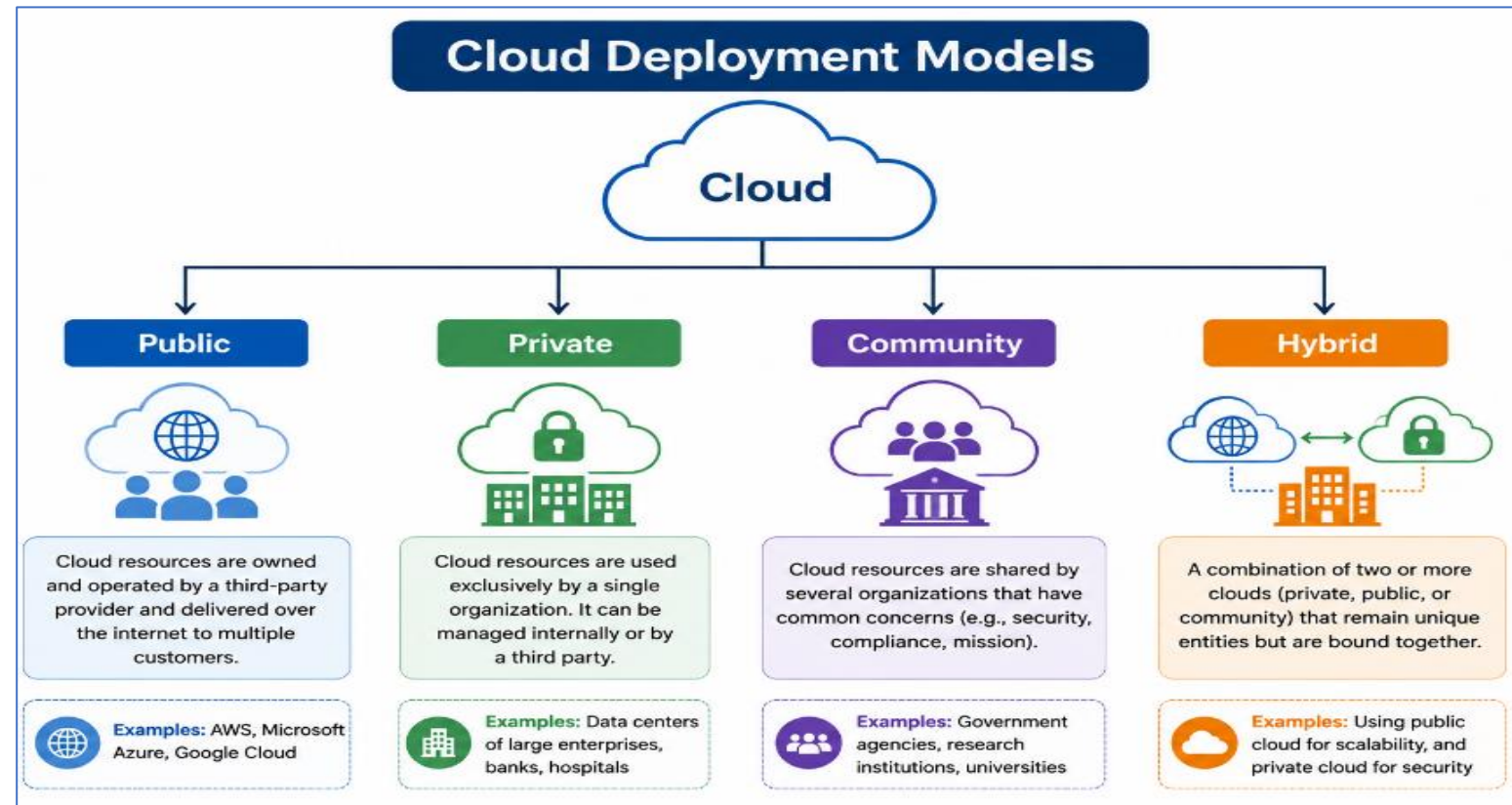
Key Points

- Four deployment models
- Ownership and access
- Business applications
- Comparison of models

CLOUD TECHNOLOGIES

Cloud Deployment Models

- The National Institute of Standards and Technology (**NIST**) defines four primary cloud deployment models:
- Public Cloud
- Private Cloud
- Community Cloud
- Hybrid Cloud



CLOUD TECHNOLOGIES

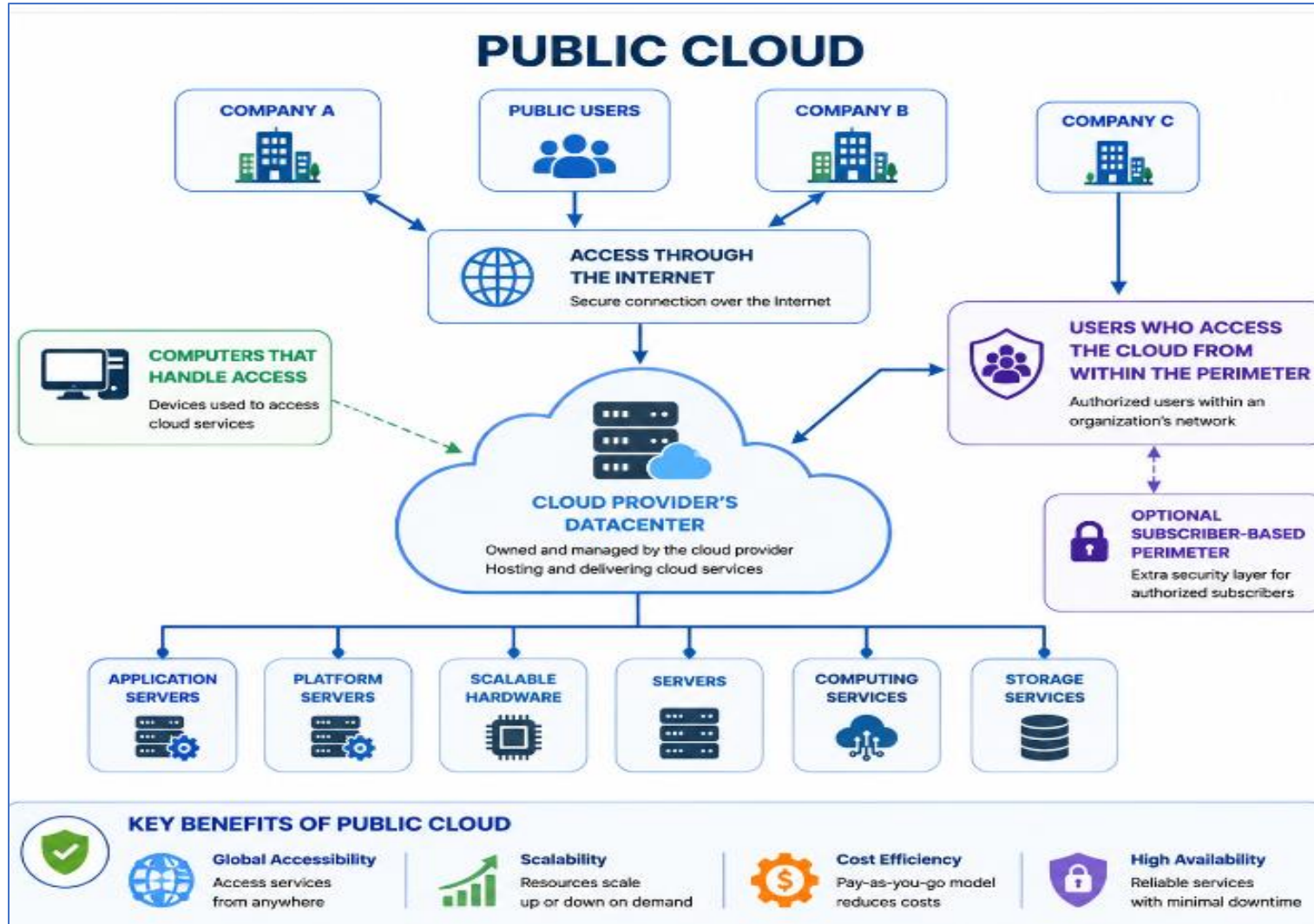
Cloud Deployment Models

- Each deployment model offers different levels of control, security,
- flexibility, and cost. Choosing the right model is essential for achieving
- business objectives while maintaining data security and operational efficiency.

- A **Public Cloud** is a cloud infrastructure owned and managed by a third-party cloud provider. Computing resources such as servers, storage, databases, and networking are shared among multiple customers over the Internet.
- Users pay only for the resources they consume using a pay-as-you-go pricing model.

CLOUD TECHNOLOGIES

Public Cloud

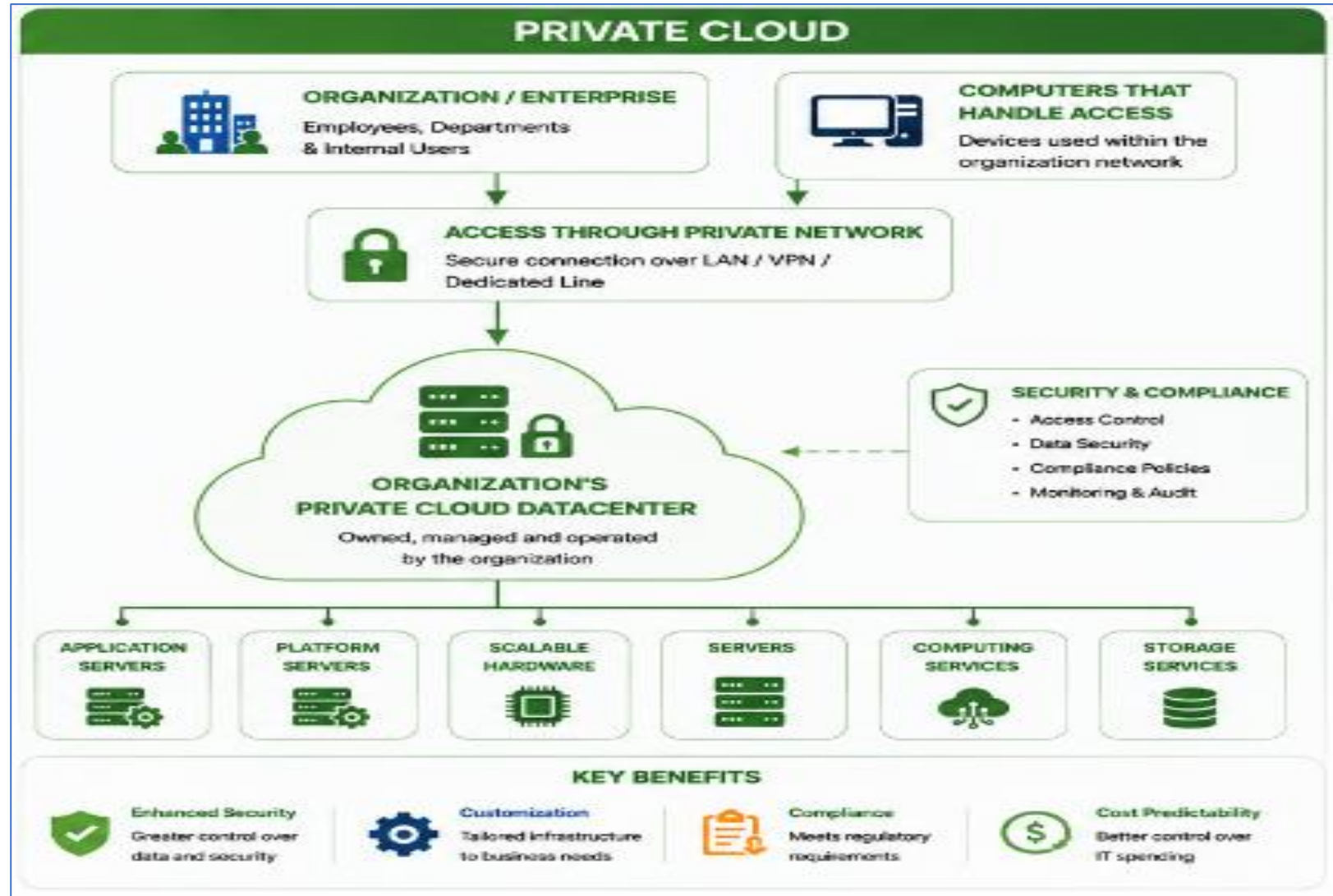


Characteristics	Advantages	Limitations	Examples
• Dedicated infrastructure • Single organization • Enhanced security • Greater control • Customizable environment	• Low initial investment • Easy deployment • Automatic updates • Global accessibility • High scalability	• Less control over infrastructure • Shared resources • May not meet strict regulatory requirements	• Amazon Web Services • Microsoft Azure • Google Cloud

- A **Private Cloud** is dedicated to a single organization. The infrastructure may be located on the organization's premises or hosted by a third-party provider, but it is used exclusively by one organization.
- Private clouds provide greater control, security, and customization, making them suitable for organizations handling sensitive information

CLOUD TECHNOLOGIES

Private Cloud



Characteristics	Advantages	Limitations	Examples
• Dedicated infrastructure • Single organization • Enhanced security • Greater control • Customizable environment	• High security • Better compliance • Full administrative control • Improved privacy • Customized infrastructure	• Higher cost • Requires skilled IT staff • Limited scalability compared to the public cloud	• Banking systems • Defence organizations • Government agencies • Large hospitals

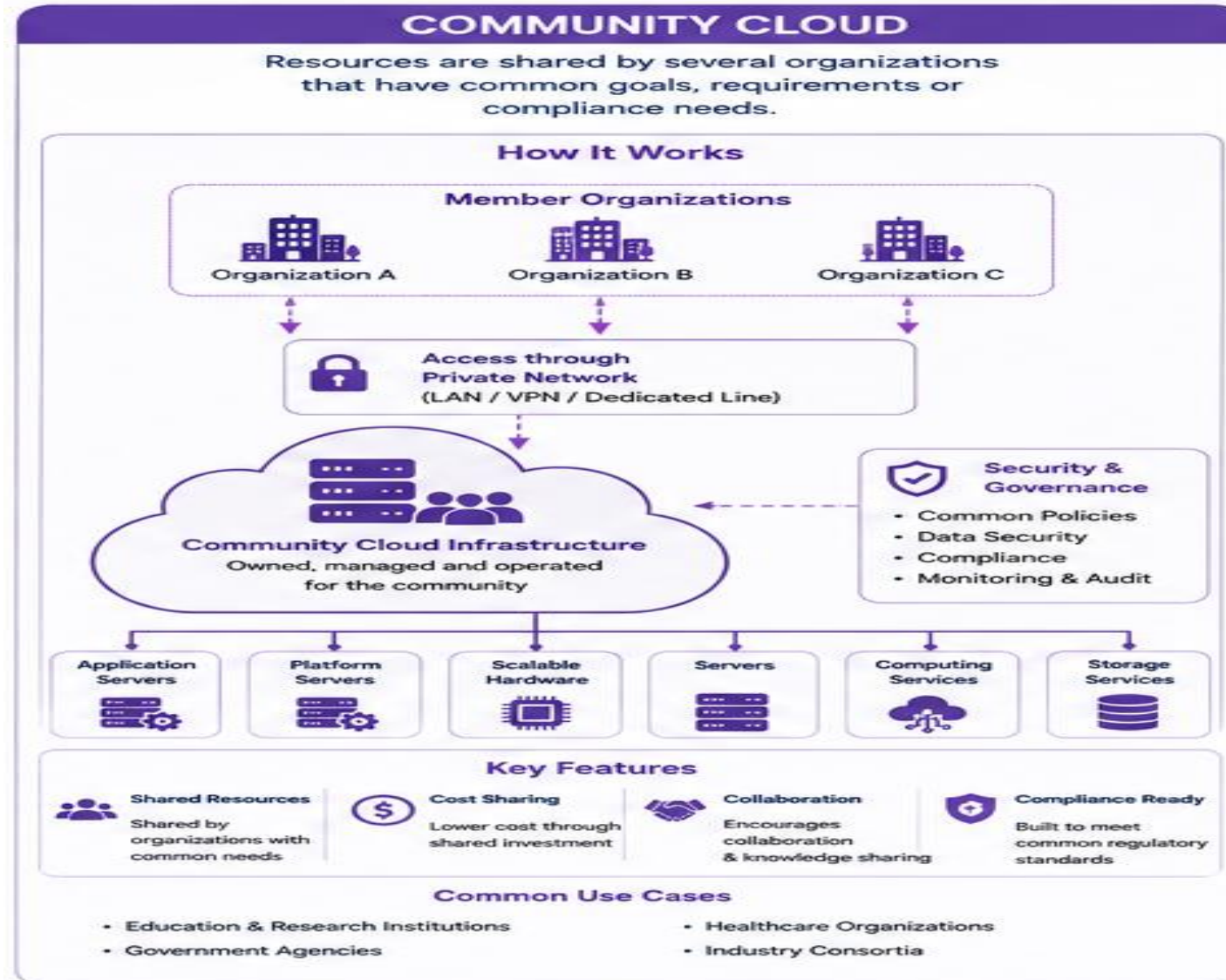
CLOUD TECHNOLOGIES

Community Cloud

- A **Community Cloud** is shared by multiple organizations with similar goals, security requirements, or regulatory needs.
- The participating organizations jointly use the infrastructure and share the associated costs.

CLOUD TECHNOLOGIES

Community Cloud

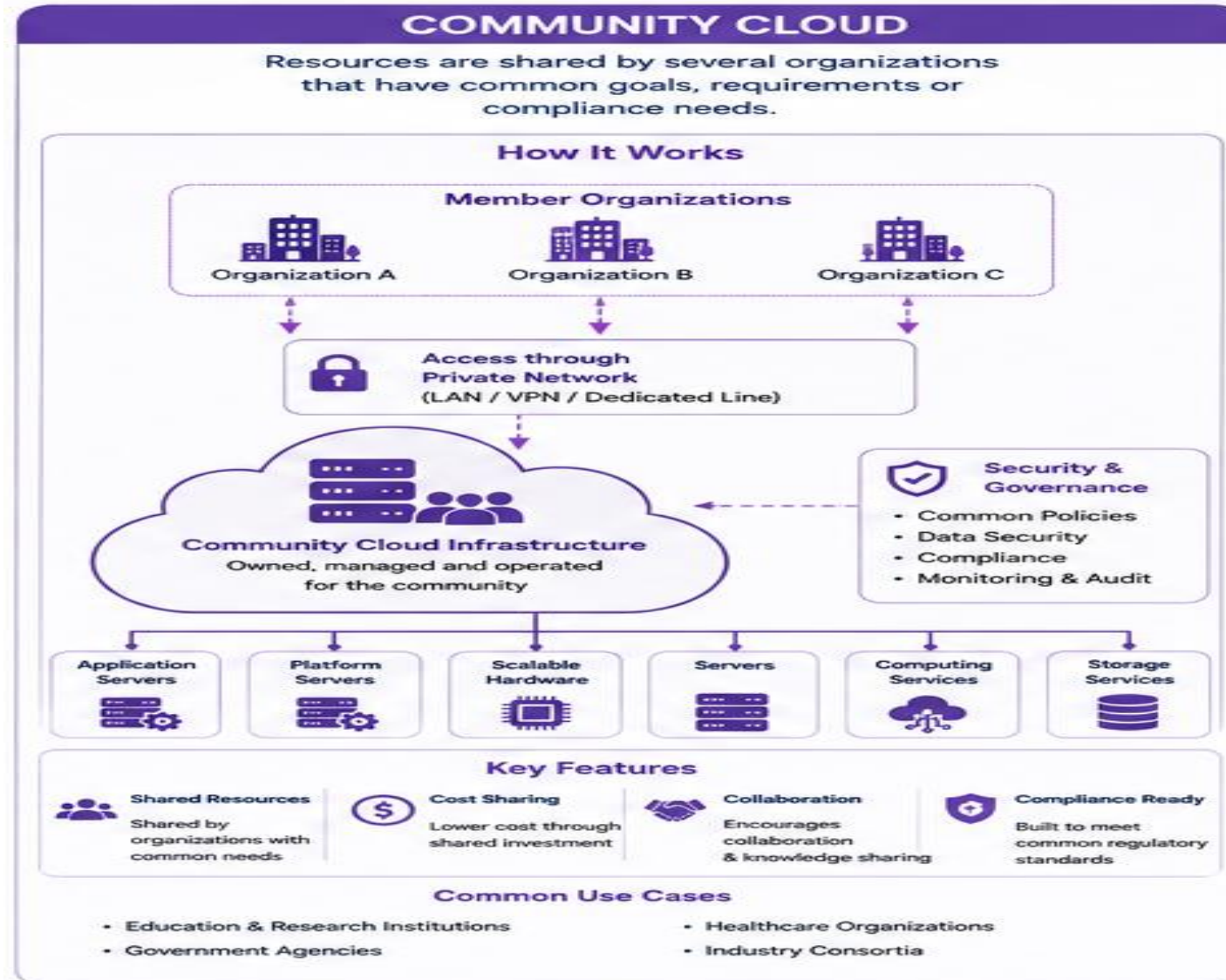


Characteristics	Advantages	Limitations	Examples
• Shared among similar organizations • Common security requirements • Shared infrastructure • Controlled access	• Lower cost than a private cloud • Better collaboration • Shared resources • Improved compliance	• Limited flexibility • Shared governance • Complex management	• Universities • Government departments • Healthcare organizations • Research institutions

- A **Hybrid Cloud** combines two or more deployment models (typically private and public clouds) connected through secure networking.
- Organizations keep sensitive applications and confidential data in the private cloud while using the public cloud for less sensitive workloads or additional computing capacity.

CLOUD TECHNOLOGIES

Hybrid Cloud



Characteristics	Advantages	Limitations	Examples
• Combination of multiple clouds • Flexible resource allocation • Enhanced security • Supports business continuity	• Best of both models • Improved flexibility • Better disaster recovery • Cost optimization • Increased scalability	• Complex integration • Higher management effort • Security configuration challenges	• Digital transformation • Disaster recovery (DR) • Development and testing (dev/test)

Session Summary

- ✓ Meaning of cloud deployment models.
- ✓ Public Cloud.
- ✓ Private Cloud.
- ✓ Community Cloud.
- ✓ Hybrid Cloud.
- ✓ Comparison of deployment models.
- ✓ Business case study on deployment model selection.

Review Questions

1. What is a cloud deployment model?
2. Explain the Public Cloud with an example.
3. Differentiate between Public Cloud and Private Cloud.
4. What is a Community Cloud?
5. Explain Hybrid Cloud with a real-world example.
6. Which deployment model is most suitable for a hospital? Why?

Question 1	Question 2	Question 3
Which deployment model is owned and managed by a cloud service provider? A. Private Cloud B. Public Cloud C. Community Cloud D. Hybrid Cloud	Which deployment model combines two or more cloud environments? A. Public Cloud B. Private Cloud C. Community Cloud D. Hybrid Cloud	Which deployment model is most suitable for organizations with strict security requirements? A. Public Cloud B. Private Cloud C. Community Cloud D. Hybrid Cloud



THANK YOU

Shivaleela B H
Computer Applications



CLOUD TECHNOLOGIES

Shivaleela B H
Computer Applications



CLOUD COMPUTING

Cloud Service Models (IaaS, PaaS, SaaS)

Shivaleela B H
Computer Applications

At the end of this session, students will be able to:

- Understand three cloud service models—Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS)
- Compare their characteristics.
- Identify suitable real-world applications.
- Choose the appropriate service model for different scenarios.

CLOUD TECHNOLOGIES

Cloud Service Models

- Cloud computing provides IT services through different service models. These models define what services the cloud provider manages and what responsibilities remain with the customer.
- Understanding these models helps organizations choose the right cloud solution based on their technical expertise, budget, and business requirements.

Key Points

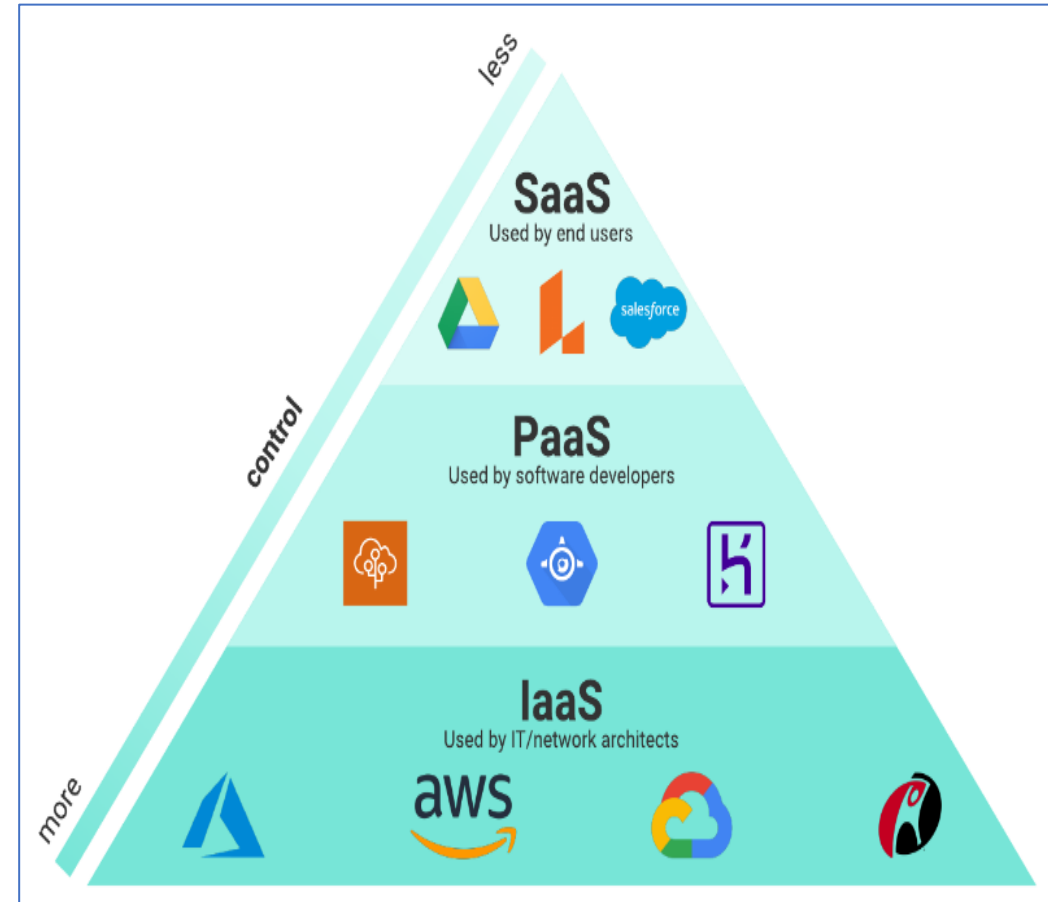
- Three service models
- Responsibilities of provider and customer
- Business applications
- Comparison

CLOUD TECHNOLOGIES

Cloud Service Models

The three primary cloud service models defined by NIST are:

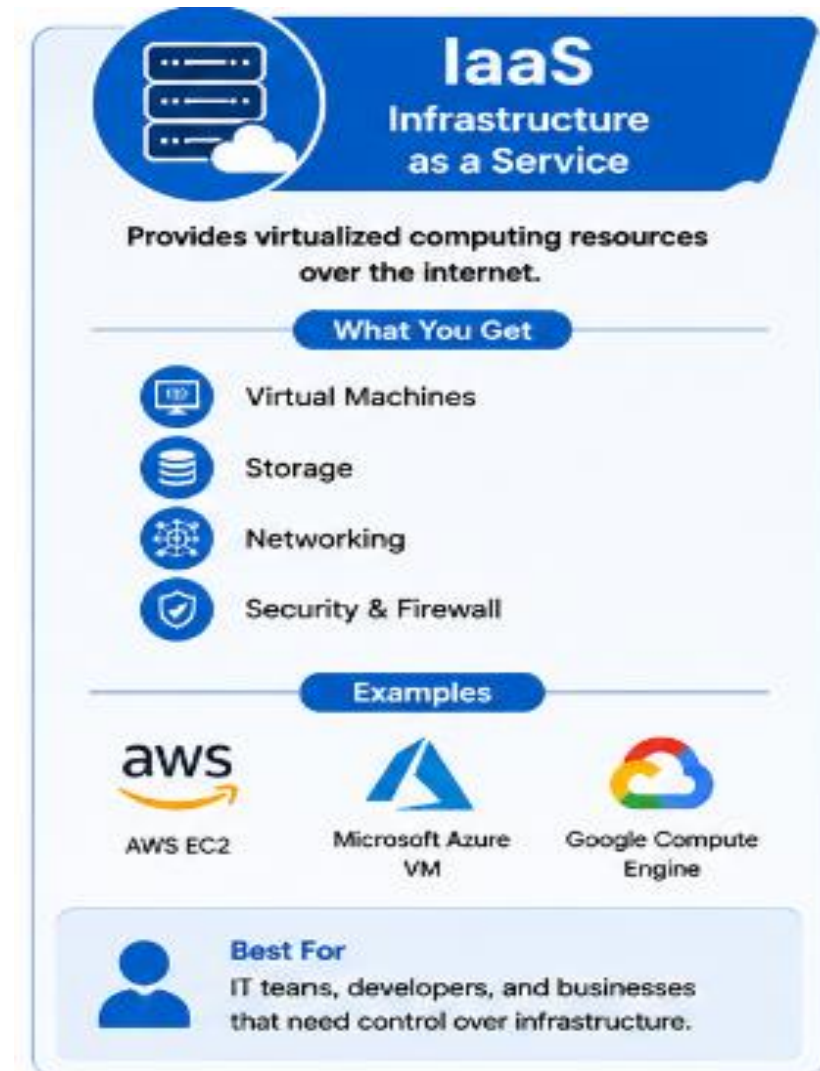
- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)
- Each model offers a different level of control and management.



CLOUD TECHNOLOGIES

Infrastructure as a Service (IaaS)

- Infrastructure as a Service (IaaS) provides virtualized computing resources such as virtual machines, storage, networking, and operating systems over the Internet.
- The cloud provider manages the physical infrastructure, while the customer installs and manages the operating system, middleware, applications, and data.

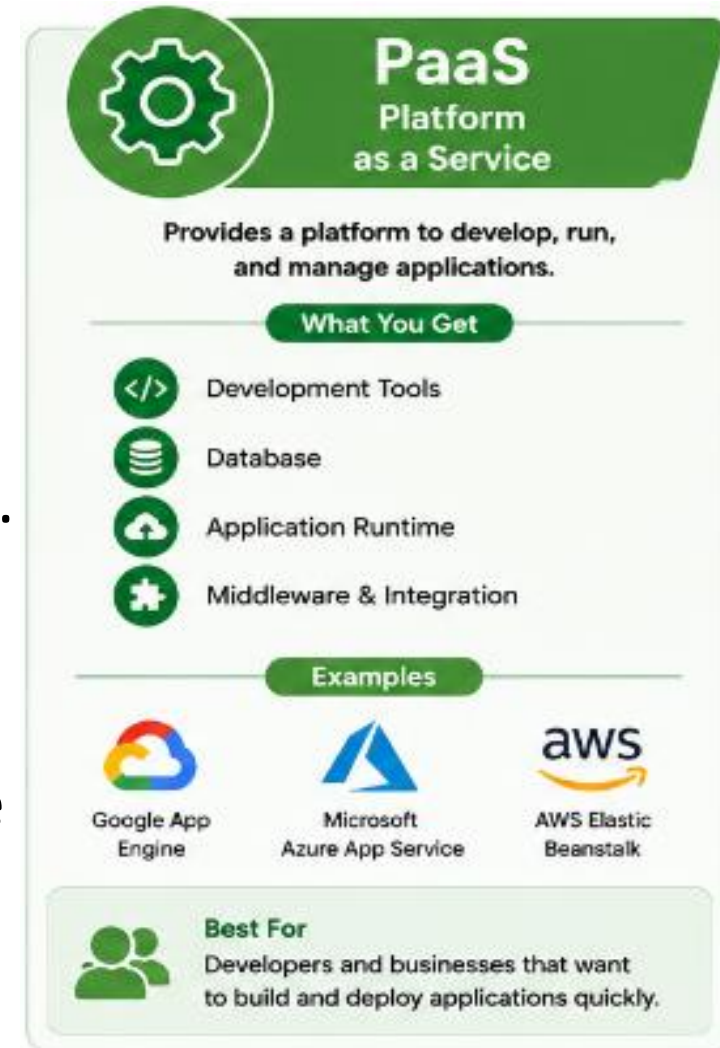


Services Provided	Advantages	Limitations	Examples
• Virtual machines • Storage • Networking • Firewalls • Load balancers	• High flexibility • Full control over software • Easy scalability • Cost-effective • Suitable for development and testing	• Requires technical expertise • Customer manages operating systems and applications	• Amazon Web Services EC2 • Microsoft Azure Virtual Machines • Google Cloud Compute Engine

CLOUD TECHNOLOGIES

Platform as a Service (PaaS)

- Platform as a Service (PaaS) provides a complete development platform that includes operating systems, programming tools, databases, middleware, and runtime environments.
- Developers focus on writing and deploying applications without worrying about infrastructure management. Applications, and data.



CLOUD TECHNOLOGIES

Platform as a Service (PaaS)

Services Provided	Advantages	Limitations	Examples
• Applications • Data • Runtime • Middleware • o/s	• Faster application development • Reduced maintenance • Automatic updates • Built-in development tools • Easy collaboration	• Less control over the underlying infrastructure • Possible vendor lock-in	• Google App Engine • Microsoft Azure App • Salesforce Platform

CLOUD TECHNOLOGIES

Software as a Service (SaaS)

- Software as a Service (SaaS) delivers ready-to-use software applications through the Internet.
Users simply sign in and start using the software without installing or maintaining it.
- The cloud provider manages everything, including infrastructure, operating systems, software updates, security, and maintenance.



CLOUD TECHNOLOGIES

Software as a Service (SaaS)

Services Provided	Advantages	Limitations	Examples
• Complete Applications • Automatic Updates • Security and Compliance • Accessible From Anywhere	• Easy to use • No installation required • Automatic updates • Accessible from anywhere • Subscription-based pricing	• Limited customization • Requires Internet access • Less control over software configuration	• Google Workspace • Microsoft Microsoft 365 • Zoom Video Communications Zoom • Dropbox

CLOUD TECHNOLOGIES

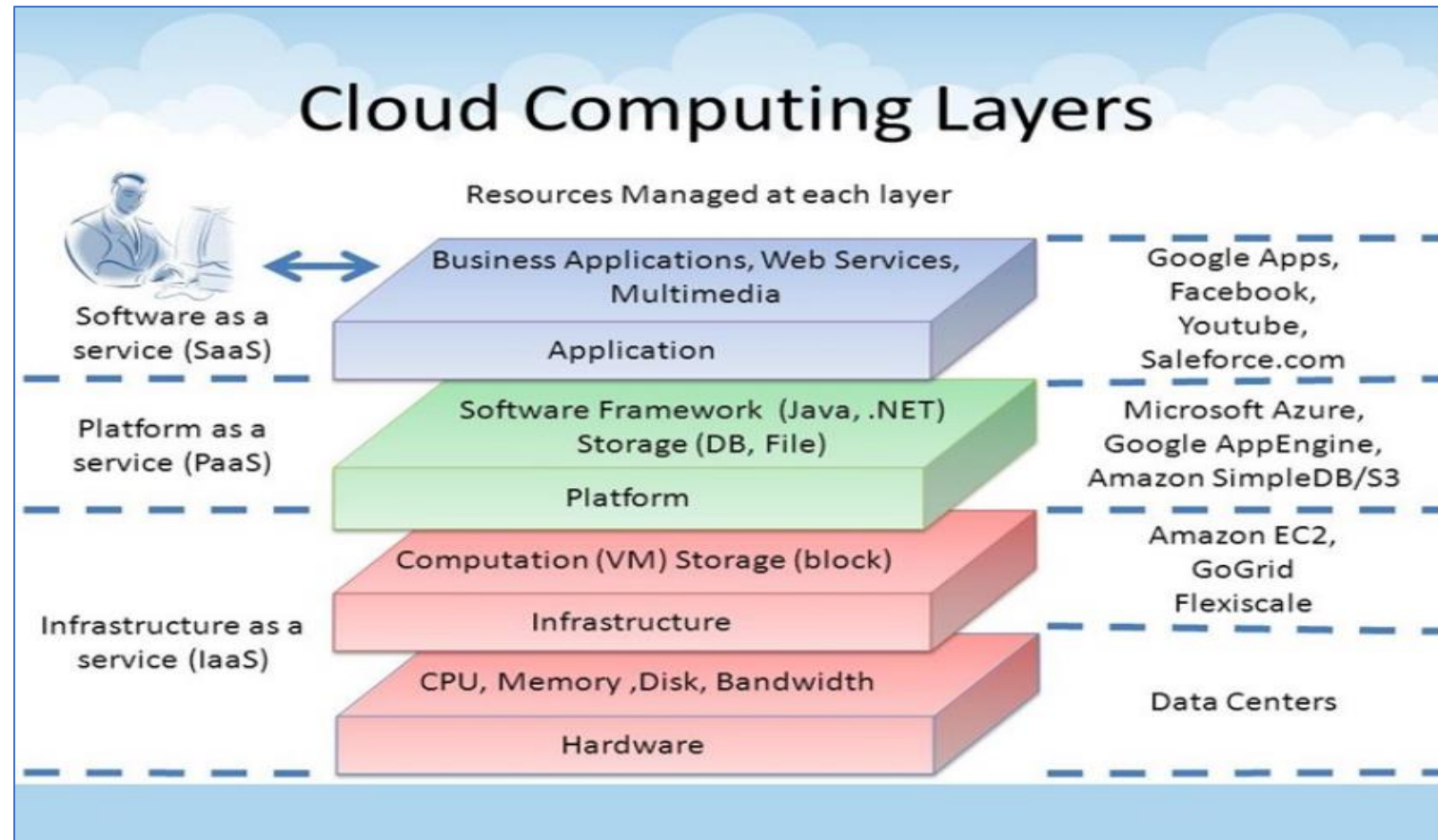
Comparison of IaaS, PaaS, and SaaS

Feature	 IaaS	 PaaS	 SaaS
Infrastructure	Provider	Provider	Provider
Operating System	Customer	Provider	Provider
Runtime	Customer	Provider	Provider
Applications	Customer	Customer	Provider
Ease of Use	Medium	High	Very High
Control	High	Medium	Low
Primary Users	IT Teams	Developers	End Users

Summary

IaaS → Maximum control, more management. **PaaS** → Balanced control and convenience.
SaaS → Ready-to-use software with minimal management.

- Cloud service models can be understood as layers, with each higher layer abstracting more of the underlying infrastructure.
- Layered Architecture
- Physical Infrastructure
- Virtualization
- Operating System
- Runtime Environment
- Middleware



CLOUD TECHNOLOGIES

Layered Architecture of Cloud Service Models

- Applications
- End Users
- The responsibility for managing these layers shifts from the customer to the cloud provider depending on the chosen service model.

CLOUD TECHNOLOGIES

Industry Examples and Use Cases

- Different organizations choose different service models based on their requirements.
- **Startup Developing a Mobile App**
 - **Service Model:** PaaS
 - **Reason:** Fast application development and deployment
- **University Using Google Classroom**
 - **Service Model:** SaaS
 - **Reason:** Ready-to-use software with minimal maintenance

CLOUD TECHNOLOGIES

Industry Examples and Use Cases

- **IT Company Hosting Its Own Applications**
 - **Service Model:** IaaS
 - **Reason:** Greater control over infrastructure and resources
- **Hospital Patient Management System**
 - **Service Model:** Hybrid Cloud (IaaS + SaaS)
 - **Reason:** Combines security with operational flexibility

- **Bank Online Services**
 - **Service Model:** IaaS + PaaS
 - **Reason:** Secure infrastructure with efficient application development

Session Summary

- ✓ Learned the three cloud service models: IaaS, PaaS, and SaaS.
- ✓ Understood the responsibilities of cloud providers and customers.
- ✓ Explored the features, benefits, and examples of each service model.
- ✓ Compared IaaS, PaaS, and SaaS based on management and usage.
- ✓ Identified real-world cloud services from AWS, Microsoft Azure, and Google Cloud.

Review Questions

1. What is Infrastructure as a Service (IaaS)?
2. Differentiate between IaaS, PaaS, and SaaS.
3. Which components are managed by the cloud provider in IaaS?
4. Give one example each of IaaS, PaaS, and SaaS.
5. Why do organizations prefer cloud service models?
6. Which cloud service model is most suitable for end users?

Question 1	Question 2	Question 3
Which cloud service provides virtual machines and storage? A. SaaS B. PaaS C. IaaS D. Community Cloud	Which cloud service provides a development platform for developers? A. IaaS B. PaaS C. SaaS D. Hybrid Cloud	Which cloud service delivers ready-to-use software over the Internet? A. IaaS B. PaaS C. SaaS D. Private Cloud



THANK YOU

Shivaleela B H
Computer Applications